

Traceability Matrix

REMEMBER GORDON! – Rulebook $\rightleftharpoons$ Implementation Mapping

Generated from `docs/traceability.toml`

Overview

Implemented	Descriptive	Implicit	Out-of-scope
86	33	2	3

Total mappings: 124 · Total impl sites: 260

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 1/3 implemented

§1 – Introduction

§1 – Introduction

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Introduction

§1.1 – General Comments

implemented
manual page 1

General Comments

“REMEMBER GORDON!” — THE BATTLE OF OMDURMAN is a simulation of the final battle in Great Britain’s two-year campaign to reassert her presence in the Sudan (1896–1898). Fought September 2nd, 1898, Omdurman finally broke the back of the fanatical Dervish rebellion and gained Britain a million square miles of desolate territory and two million impoverished subjects. With two players, one assumes the role of Herbert Kitchener, Sirdar (CIC) of the Anglo-Egyptian army; the other player becomes the Khalifa, Abdullah the Taiasha, absolute ruler of the Dervishes. The game is also suited for multi-player participation, with each player assuming command of one or more Dervish tribes or Anglo-Egyptian brigades.

While “REMEMBER GORDON!” — THE BATTLE OF OMDURMAN is not, strictly speaking, a beginner’s game, the mechanics of play should be familiar to players of modest experience. It is suggested that the bonus game, FALL OF KHARTOUM, and the historical scenario be played first to familiarize players with the game system prior (see manual for full text)

File	Symbol	Code Snippet
omdurman-rules/src/unit_profiles.rs:194 GH:omdurman-rules/src/unit_profiles.rs:194	command_owns_unit	<pre>192 /// treat *nobody-claims-it* units as the faction's communal pool, so `false` 193 /// here means "any member of the faction may act on this unit". 194 pub fn command_owns_unit(scope: &omdurman_types::CommandScope, identity: &UnitIdentity) -> bool { 195 match scope { 196 omdurman_types::CommandScope::Army => false,</pre>

Covered by tests: [omdurman-rules::src::unit_profiles::command_owns_unit_follows_tribes_and_brigades](#)
[omdurman-net::src::lib::start_game_commands_round_trip](#)
[omdurman-net::src::lib::legacy_start_game_without_commands_still_loads](#)
[omdurman-net::src::lib::brigade_scope_display_and_membership](#)
[omdurman-app::src::tests::replayed_start_game_stages_commands_and_resets_ready](#)

§1.2 – Game Scale

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Game Scale

Each hexagon of the mapsheet represents approximately 400–440 yards of real terrain and each day turn is the equivalent of two hours of real time. Each counter of infantry and cavalry represents between 400 and 700 men, and each of the gunboats present at the battle has its own counter. The upper echelon of command is represented by individual leader counters for the Anglo-Egyptian force; and leaders plus their retinues for the Dervish army.

§2 – Game Components

§2 – Game Components

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Game Components

§2.1 – The Game Maps

descriptive

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The Game Maps

The Omdurman battle map represents approximately 100 square miles of real territory and includes the area north of Omdurman in which the historical battle took place as well as the dominant terrain features that influenced the course of the battle. Note that the mapsheet also contains the Turn Record Track, Turn Sequence, and Terrain Effects Chart at the top; and the Combat Tables and Howitzer Fire Scattergram in the lower right corner. The large letters “A”, “D”, “Y”, etc. are set-up hexes for the historical scenario only (9.2) and should be ignored in the campaign game. Similarly, the hexsides of the Zariba exist only in the historical scenario and should be considered clear terrain in the campaign game. Note, however, that the Anglo-Egyptian player may “construct” the Zariba in the campaign game if desired (see 5.3). All full hexes of the Omdurman game map are playable, including the seven hexes of the Howitzer Fire Scattergram.

The mini-map for the bonus game, FALL OF KHARTOUM, represents that city as it appeared in 1885. The portion (see manual for full text)

See also: §5.3, §9.2

§2.2 – Play Aids

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Play Aids

Certain charts and tables are needed to play the game. The Terrain Effects Chart lists all terrain found on the mapsheet and the effect of each type on movement and combat. The Combat Tables describe the range effects on various weapon types and includes the Combat Results Table. Also note the Line of Sight Table on the back of this rulebook. It tells players when certain terrain types block line of sight, thus preventing direct fire attacks on enemy units. Players should become familiar with these various charts and tables prior to the beginning of play.

(see manual for full text)

§2.3 – The Units

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The Units

File	Symbol	Code Snippet
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omdurman-types/src/lib.rs:1043 GH:omdurman-types/src/lib.rs:1043	BrigadeId	<pre> 1041 /// same field for uniform handling. 1042 #[derive(Serialize, Deserialize, Clone, Copy, PartialEq, Eq, PartialOrd, Ord, Hash, Debug)] 1043 pub struct BrigadeId { 1044 pub number: u8, 1045 pub nationality: BrigadeNationality,</pre>
omdurman-rules/src/unit.rs:491 GH:omdurman-rules/src/unit.rs:491	UnitProfile	<pre> 489 /// print no melee value). 490 #[derive(Serialize, Deserialize, Clone, Copy, PartialEq, Eq, Debug)] 491 pub struct UnitProfile { 492 pub kind: UnitKind, 493 pub identity: UnitIdentity,</pre>
omdurman-rules/src/lib.rs:14 GH:omdurman-rules/src/lib.rs:14	BrigadeId	<pre> 12 13 use omdurman_types::{ 14 BrigadeId, BrigadeNationality, DayNight, DervishTribe, HexCoord, Player, UnitKind, 15 }; 16 </pre>
omdurman-types/src/lib.rs:679 GH:omdurman-types/src/lib.rs:679	SpriteAnnotation	<pre> 677 /// as an optional overlay over the compiled `sprite_data` fallback. 678 #[derive(Serialize, Deserialize, Clone, Debug, PartialEq)] 679 pub struct SpriteAnnotation { 680 pub color: SpriteColor, 681 pub faction: Option<Faction>,</pre>

Covered by tests: [omdurman-rules::src::unit_profiles::british_army_row_zero_specials_classify_by_counter](#)
[omdurman-rules::src::unit_profiles::egyptian_army_row_zero_specials_classify_by_counter](#)
[omdurman-rules::src::unit_profiles::tribe_stats_come_from_annotation](#)
[omdurman-rules::src::unit_profiles::section_owner_dervish_sections](#)
[omdurman-rules::src::unit_profiles::section_owner_green_sections_are_dervish](#)

§2.4 – Game Parts Inventory

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Game Parts Inventory

Your complete copy of “REMEMBER GORDON!” — THE BATTLE OF OMDURMAN includes:

- One 22 × 28 Battle of Omdurman mapsheet
- One 8½ × 11 bonus game: FALL OF KHARTOUM mapsheet
- One Rules Booklet
- One die-cut Unit Counter Sheet
- One Campaign Game Order of Appearance Card
- One ten-sided die
- One game box

§2.31 – Dervish weapon types

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Dervish artillery, gunboats, and forts fire on the “artillery” line of the Dervish Range Effects Table; Jehadia and Danagla units fire on the “rifles” line as does the Isa Zachneih unit. All other Dervish units (including leaders) are armed with spears and swords.

File	Symbol	Code Snippet
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omdurman-rules/src/unit.rs:167 GH:omdurman-rules/src/unit.rs:167	WeaponClass	<pre> 165 Serialize, Deserialize, Clone, Copy, PartialEq, Eq, PartialOrd, Ord, Hash, Debug, strum::Display, 166)] 167 pub enum WeaponClass { 168 /// Dervish spears and swords -- no ranged fire at all. 169 Melee, </pre>
omdurman-rules/src/unit_profiles.rs:578 GH:omdurman-rules/src/unit_profiles.rs:578	dervish_tribe	<pre> 576 /// Resolve a Dervish tribal foot counter (§2.31): Jehadia, Danagla and 577 /// Isa Zachneih fire on the rifles line; every other tribe is spear-armed. 578 pub fn dervish_tribe(tribe: DervishTribe) -> Option<Classification> { 579 // §2.31: "Jehadia and Danagla units fire on the 'rifles' line as does the 580 // Isa Zachneih unit. All other Dervish units (including leaders) are armed </pre>
omdurman-rules/src/unit_profiles.rs:344 GH:omdurman-rules/src/unit_profiles.rs:344	khalifa_abdullah	<pre> 342 /// battle (§9.322). All three are interchangeable, so they share the 343 /// `DervishArtillery` identity. 344 pub fn khalifa_abdullah(col: u32, row: u32) -> Option<Classification> { 345 let artillery = { 346 Some(Classification { </pre>

Covered by tests: [omdurman-rules::src::unit_profiles::dervish_weapon_class_follows_the_rifles_line](#)

§2.32 – Anglo-Egyptian weapon types

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All Anglo-Egyptian units (except gunboats, Maxims, artillery, and leaders) are armed with rifles. Maxims fire on the “Maxims” line of the Anglo-Egyptian Range Effects Table, and artillery and old gunboats fire on the “Artillery” line. New type (named) gunboats may fire on the “Howitzer”, “Artillery”, and “Maxims” lines of the Range Effects Table. (See 6.52 for the fire capabilities of the “Friendlies”).

****Sample Dervish Units**** (printed on counters): combat unit (Combat / Melee / Movement values, plus Tribe identifier); Leader (e.g. OSMAN DIGNA); Camel unit (e.g. Danagla, 4-6-12); Fort.

****Sample Anglo-Egyptian Units**** (printed on counters): Cavalry (e.g. 21 Lancers); Artillery (e.g. 32 Battery); Old Gunboat (e.g. LORD KITCHENER, 0-0-15); New Gunboat — named (e.g. Sultan, with artillery and howitzer factor, plus movement downstream / movement upstream values); Maxim Guns (fire twice per turn); Infantry (Fire Combat Factor / Melee / Movement, plus Battalion ID and Brigade ID — e.g. “2B” = 2nd British Brigade, “3E” = 3rd Egyptian Brigade).

(see manual for full text)

See also: §6.52

File	Symbol	Code Snippet
omdurman-rules/src/unit.rs:119 GH:omdurman-rules/src/unit.rs:119	GunboatId	<pre> 117 /// fire; "old" gunboats do not (rulebook §2.32). 118 #[derive(Serialize, Deserialize, Clone, Copy, PartialEq, Eq, Hash, Debug, strum::Display)] 119 pub enum GunboatId { 120 /// One of the five new-type named gunboats with howitzer capability. 121 Named(NamedGunboat), </pre>
omdurman-rules/src/unit.rs:138 GH:omdurman-rules/src/unit.rs:138	NamedGunboat	<pre> 136 /// The five named gunboats with howitzer capability (rulebook §6.64, §2.32). 137 #[derive(Serialize, Deserialize, Clone, Copy, PartialEq, Eq, Hash, Debug, strum::Display)] 138 pub enum NamedGunboat { 139 Sultan, 140 Melik, </pre>

omdurman-rules/src/unit.rs:151 GH:omdurman-rules/src/unit.rs:151	OldGunboat	<pre>149 /// in the Maxim Second Fire and Howitzer subphase (\$6.42). 150 #[derive(Serialize, Deserialize, Clone, Copy, PartialEq, Eq, Hash, Debug, strum::Display)] 151 pub enum OldGunboat { 152 LordKitchener, 153 Tamai,</pre>
omdurman-rules/src/unit.rs:151 GH:omdurman-rules/src/unit.rs:151	GunboatId::Old	<pre>149 /// in the Maxim Second Fire and Howitzer subphase (\$6.42). 150 #[derive(Serialize, Deserialize, Clone, Copy, PartialEq, Eq, Hash, Debug, strum::Display)] 151 pub enum OldGunboat { 152 LordKitchener, 153 Tamai,</pre>
omdurman-rules/src/unit.rs:125 GH:omdurman-rules/src/unit.rs:125	GunboatId::Dervish-Gunboat	<pre>123 Old(OldGunboat), 124 /// A Dervish gunboat (§9.111, §10.14). 125 DervishGunboat(u8), 126 } 127 </pre>

Covered

by

tests:

omdurman-rules::src::effects::tests::old_gunboat_lacks_howitzer

omdurman-rules::src::effects::tests::old_gunboat_rejected_from_howitzer_subphase

0/1 implemented

§3 – Getting Started

§3 – Getting Started

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Getting Started

Spread out the mapsheet on a table. It should lie flat if you backfold it against the scored lines. The Dervish player should sit next to the west edge of the map and the Anglo-Egyptian player opposite him on the east edge. Read through the rules once, looking over the various charts as they are referred to in the various sections. Next, select a scenario and punch out only those unit counters needed to play. Later on, the rest of the unit counters should be punched out, sorted and stored by unit type.

(see manual for full text)

■ 1/1 implemented

§4 – Turn Sequence

§4 – Turn Sequence

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Turn Sequence

“REMEMBER GORDON!” — THE BATTLE OF OMDURMAN is played in “Game Turns”, each of which has two “Player Turns”. The player moving first will vary according to the scenario being played. In the Campaign Game, for example, the Anglo-Egyptian player moves first.

A) Anglo-Egyptian Player Turn:

1. Anglo-Egyptian Movement Phase
2. Fire Combat Phase a. Dervish Defensive Fire b. Anglo-Egyptian Offensive Fire
 1. Direct Fire Subphase
 2. Maxim Second Fire and Howitzer Fire Subphase

(see manual for full text)

File	Symbol	Code Snippet
omdurman-rules/src/scalars.rs:184 GH:omdurman-rules/src/scalars.rs:184	GameTurnIndex	<pre>182 /// One-based Game Turn index (1, 2, ... up to the 183 scenario length) (rulebook §4). 184 #[derive(Serialize, Deserialize, Clone, Copy, 185 PartialEq, Eq, PartialOrd, Ord, Hash, Debug)] 186 pub struct GameTurnIndex(pub(crate) u8); 187 188 impl GameTurnIndex {</pre>
omdurman-rules/src/turn.rs:16 GH:omdurman-rules/src/turn.rs:16	Phase	<pre>14 /// etc. 15 #[derive(Serialize, Deserialize, Clone, Copy, 16 PartialEq, Eq, Hash, Debug, Default)] 17 pub enum Phase { 18 /// Pre-game deployment (§9.2/§9.3/§10): fixed 19 /// units are placed, each side 20 /// deploys its order of battle within its legal 21 /// zone, and river</pre>
omdurman-rules/src/effects/state.rs:21 GH:omdurman-rules/src/effects/state.rs:21	GameState	<pre>19 /// All mutable state of a game in progress (rulebook 20 §4). 21 #[derive(Serialize, Deserialize, Clone, Debug)] 22 pub struct GameState { 23 pub scenario: Scenario, 24 pub current_turn: GameTurnIndex,</pre>
omdurman-rules/src/effects/state.rs:172 GH:omdurman-rules/src/effects/state.rs:172	GameState::new	<pre>170 impl GameState { 171 /// Create a fresh game state for a given scenario 172 /// (rulebook §4). 173 pub fn new(scenario: Scenario) -> Self { 174 let first = scenario_turn(scenario, 175 GameTurnIndex::new(1)); 176 // First player to *move* per scenario: 177 Campaign -- Anglo-Egyptian moves</pre>
omdurman-rules/src/effects/effect.rs:17 GH:omdurman-rules/src/effects/effect.rs:17	AdvancePhase	<pre>15 /// At end-of-turn, disrupted units recover and 16 /// per-turn tracking is 17 /// cleared. 18 AdvancePhase, 19 // -- Movement 20 -----</pre>
omdurman-rules/src/effects/dispatch.rs:192 GH:omdurman-	advance_phase	<pre>190 191 /// Advance the game state to the next phase (rulebook 192 §4).</pre>

rules/src/effects/ dispatch.rs:192		<pre>192 pub fn advance_phase(state: &mut GameState) -> Result<(), RuleError> { 193 let old_phase = state.phase; 194 #[cfg(not(feature = "kani"))]</pre>
omdurman- rules/src/effects/ dispatch.rs:298 GH:omdurman- rules/src/effects/ dispatch.rs:298	end_player_turn	<pre>296 297 /// End the current player's turn: recover disrupted units, switch active player, advance turn index (rulebook §4). 298 pub fn end_player_turn(state: &mut GameState) { 299 #[cfg(not(feature = "kani"))] 300 debug!()</pre>
omdurman- rules/src/ scalars.rs:190 GH:omdurman- rules/src/ scalars.rs:190	GameTurnIn- dex::value	<pre>188 Self(value) 189 } 190 pub fn value(self) -> u8 { 191 self.0 192 }</pre>
omdurman- rules/src/effects/ state.rs:164 GH:omdurman- rules/src/effects/ state.rs:164	PendingMelee	<pre>162 /// resolution after the reaction window is deterministic and host-ordered (rulebook §7.5). 163 #[derive(Serialize, Deserialize, Clone, Debug)] 164 pub struct PendingMelee { 165 pub attack: MeleeAttack, 166 pub attacker_roll: DieRoll,</pre>

Covered by tests: omdurman-rules::src::effects::tests::both_ready_auto_advances_out_of_setup
omdurman-rules::src::effects::tests::fire_combat_wrong_phase_rejected omdurman-rules::src::effects::tests::new_game_starts_in_setup
omdurman-rules::src::turn_track::scenario_turn_dispatches_correctly
omdurman-rules::src::effects::tests::turn_advances_through_phases
omdurman-rules::src::unit_profiles::game_turn_marker_cell_returns_none
omdurman-rules::src::effects::tests::declared_melee_blocks_phase_advance

18/23 implemented

§5 – Movement Phase

§5 – Movement Phase (general)

descriptive

manual page unknown

Movement Phase

§5.1 – General Rules

descriptive

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General Rules

§5.2 – Movement Restrictions

descriptive

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Movement Restrictions

§5.3 – Constructing the Zariba

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Constructing the Zariba

The Zariba trench and thorn hedge hexsides are built and in place in the historical scenario only. These hexsides are considered clear terrain in the campaign game. The Anglo-Egyptian player may, however, find it useful to construct this defensive position during the campaign game. The Zariba hexsides may only be built in their position as displayed on the mapsheet. Construction procedure is as follows: any Anglo-Egyptian infantry unit that begins and ends the Anglo-Egyptian player turn adjacent to (and on the Nile side of) Zariba hexsides has constructed all Zariba hexsides to which he is adjacent. The constructing unit may neither fire offensively nor melee attack during the turn of construction. Use a blank counter to denote units constructing Zariba hexsides. See 9.23 for defensive benefits and movement restrictions of Zariba hexsides.

(see manual for full text)

See also: §9.23

File	Symbol	Code Snippet
omdurman-rules/src/unit.rs:525 GH:omdurman-rules/src/unit.rs:525	constructing_zariba	<pre>523 /// Set while the unit is building Zariba hexsides -- -- neither offensive 524 /// fire nor melee allowed that turn (§5.3). 525 pub constructing_zariba: bool, 526 /// Set when the Royal Engineers are committed to a demolition this turn 527 /// (§6.53) -- neither offensive fire nor melee allowed that turn.</pre>
omdurman-rules/src/effects/effect.rs:170 GH:omdurman-	ConstructZariba	<pre>168 169 /// Begin constructing a Zariba hexside (rulebook §5.3). 170 ConstructZariba {</pre>

rules/src/effects/ effect.rs:170		<pre> 171 unit_ids: Vec<UnitId>, 172 hexside: HexsideRef, </pre>
omdurman-rules/src/effects/dispatch.rs:49 GH:omdurman-rules/src/effects/dispatch.rs:49	apply_construct_zariba	<pre> 47 GameEffect::RecoverUnit { unit_id } => apply_recover_unit(state, *unit_id), 48 GameEffect::ConstructZariba { unit_ids, hexside } => { 49 apply_construct_zariba(state, unit_ids, *hexside) 50 } 51 GameEffect::Demolition { unit_id, target } => apply_demolition(state, *unit_id, *target), </pre>

Covered by tests: [omdurman-rules::src::effects::tests::construct_zariba_marks_builders_and_records_hexside](#)
[omdurman-rules::src::effects::tests::zariba_builders_may_not_fire_offensively_or_melee_until_turn_end](#)
[omdurman-rules::src::effects::tests::zariba_construction_is_a_campaign_movement_phase_action](#)
[omdurman-rules::src::effects::tests::world_lens_composes_breaches_and_constructed_zariba](#)

§5.4 – Zones of Control

[descriptive](#)

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Zones of Control

§5.5 – Stacking

[descriptive](#)

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Stacking

§5.11 – Movement allowances printed on units

[implemented](#)

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The movement allowances of the various unit types are printed directly on the units (see 2.3). A unit may move up to this printed movement allowance, paying varying costs for different terrain types (see the Terrain Effects Chart).

See also: §2.3

File	Symbol	Code Snippet
omdurman-rules/src/scalars.rs:63 GH:omdurman-rules/src/scalars.rs:63	MovementAllowance	<pre> 61 /// is a named variant. 62 #[derive(Serialize, Deserialize, Clone, Copy, PartialEq, Eq, Hash, Debug)] 63 pub enum MovementAllowance { 64 /// Immobile (forts, wrecked gunboats). 65 Immobile = 0, </pre>
omdurman-rules/src/unit.rs:502 GH:omdurman-rules/src/unit.rs:502	UnitMovement	<pre> 500 /// Movement allowance -- uniform for land units, split for gunboats (rulebook §5.11, §5.24, §5.25). 501 #[derive(Serialize, Deserialize, Clone, Copy, PartialEq, Eq, Debug)] 502 pub enum UnitMovement { 503 Land(MovementAllowance), 504 Gunboat(GunboatMovement), </pre>
omdurman-rules/src/scalars.rs:105 GH:omdurman-	MovementPoints	<pre> 103 Serialize, Deserialize, Clone, Copy, PartialEq, Eq, PartialOrd, Ord, Hash, Debug, Default, 104)] 105 pub struct MovementPoints(pub(crate) i16); </pre>

rules/src/ scalars.rs:105		106 107 <code>impl MovementPoints {</code>
omdurman- rules/src/terrain_chart.rs:21 GH:omdurman- rules/src/ terrain_chart.rs:21	terrain_effects_chart	19 /// 20 /// Source: printed Terrain Effects Chart on the mapsheet. 21 <code>pub fn terrain_effects_chart(terrain: Terrain) -></code> <code>TerrainEntry {</code> 22 <code> match terrain {</code> 23 <code> Terrain::Clear { .. } => TerrainEntry {</code>
omdurman- rules/src/terrain_chart.rs:77 GH:omdurman- rules/src/ terrain_chart.rs:77	movement_cost	75 /// Convenience: get the movement cost for a terrain type (rulebook §5.11, Terrain Effects Chart). 76 /// Returns `None` for impassable terrain (Nile). 77 <code>pub fn movement_cost(terrain: Terrain) -></code> <code>Option<MovementAllowance> {</code> 78 <code> terrain_effects_chart(terrain).movement_cost</code> 79 <code>}</code>
omdurman- rules/src/terrain_chart.rs:85 GH:omdurman- rules/src/ terrain_chart.rs:85	movement_cost_with_road	83 /// underlying terrain; without a road it's the terrain's own cost. The road is 84 /// a movement overlay only -- combat/LOS still use the underlying terrain. 85 <code>pub fn movement_cost_with_road(terrain: Terrain, road:</code> <code>bool) -> Option<MovementAllowance> {</code> 86 <code> if road {</code> 87 <code> Some(MovementAllowance::One)</code>
omdurman- types/src/ lib.rs:498 GH:omdurman- types/src/lib.rs:498	Terrain::has_road	496 497 <code> /// Whether this hex has any road touching it.</code> 498 <code>pub fn has_road(self) -> bool {</code> 499 <code> !matches!(self.road(), Road::None)</code> 500 <code>}</code>
omdurman- types/src/ lib.rs:435 GH:omdurman- types/src/lib.rs:435	Terrain::passable_by_land	433 434 <code> /// Whether this terrain may be entered by land</code> <code>units (rulebook §5.11).</code> 435 <code>pub fn passable_by_land(self) -> bool {</code> 436 <code> !self.is_nile()</code> 437 <code>}</code>
omdurman- types/src/ lib.rs:503 GH:omdurman- types/src/lib.rs:503	Terrain::is_crossroad	501 502 <code> /// Whether roads converge at this hex's centre.</code> 503 <code>pub fn is_crossroad(self) -> bool {</code> 504 <code> matches!(self.road(), Road::Crossroad)</code> 505 <code>}</code>
omdurman- rules/src/ board.rs:342 GH:omdurman- rules/src/ board.rs:342	BoardInfo::roads	340 <code> tiles: tiles.into_iter().collect(),</code> 341 <code> hexsides: Vec::new(),</code> 342 <code> roads: Vec::new(),</code> 343 <code> excluded: BTreeSet::new(),</code> 344 <code> overlay: default_overlay(),</code>

Proven by: `omdurman-rules::src::terrain_chart::movement_column_matches_the_printed_chart`
`omdurman-types::src::lib::terrain_road_state_round_trips`

Covered by tests: `omdurman-rules::src::effects::tests::extreme_coordinates_are_rejected_without_panicking`
`omdurman-rules::src::effects::tests::move_cost_is_engine_computed_and_caller_cost_ignored`
`omdurman-rules::src::effects::tests::move_path_must_be_contiguous_and_end_at_destination`
`omdurman-rules::src::terrain_chart::clear_terrain_no_bonus`
`omdurman-rules::src::terrain_chart::nile_is_impassable` `omdurman-rules::src::terrain_chart::rough_movement_and_defense`
`omdurman-rules::src::terrain_chart::swamp_movement_and_defense`
`omdurman-rules::src::terrain_chart::hilltop_movement_and_defense`
`omdurman-rules::src::terrain_chart::huts_movement_and_defense`
`omdurman-rules::src::terrain_chart::movement_cost_convenience_matches_chart`
`omdurman-rules::src::terrain_chart::movement_cost_with_road_always_one`
`omdurman-rules::src::effects::tests::land_unit_may_not_enter_nile`
`omdurman-rules::src::terrain_chart::movement_cost_without_road_matches_terrain`
`omdurman-rules::src::effects::tests::movement_cost_for_uses_terrain`

omdurman-rules::src::effects::tests::movement_cost_for_road_costs_one
omdurman-rules::src::terrain_chart::road_gives_crossroad omdurman-rules::src::terrain_chart::terrain_movement_costs_in_bounds
omdurman-rules::src::terrain_chart::terrain_chart_road_always_costs_one
omdurman-rules::src::terrain_chart::trees_cost_one_and_give_no_defense

§5.12 – Move up to allowance, hex by hex (cumulative MP cap)

implemented

manual page unknown

A player may move as many or as few of his units as desired during each movement phase, limited only by the units' movement allowance, the terrain costs paid in moving from hex to hex, and enemy zones of control (see 5.4).

See also: §5.4

File	Symbol	Code Snippet
omdurman-rules/src/ef-fects/state/movement.rs:25 GH:omdurman-rules/src/ef-fects/state/movement.rs:25	can_move_unit	<pre>23 /// Returns the same `RuleError` the `MoveUnit` effect would on rejection. 24 /// Lets the UI gate input without mutating or duplicating the rules. 25 pub fn can_move_unit(&self, unit_id: UnitId, cost: MovementPoints) -> Result<(), RuleError> { 26 let unit = self.unit_or_err(unit_id)?; 27 self.movement_preconditions(unit)?;</pre>
omdurman-rules/src/effects/state.rs:382 GH:omdurman-rules/src/effects/state.rs:382	mp_spent	<pre>380 381 /// Movement points `unit_id` has already spent this turn (§5.11/§5.12). 382 pub fn mp_spent(&self, unit_id: UnitId) -> i16 { 383 self.mp_spent_this_turn.get(&unit_id).copied().unwrap_or(0) 384 }</pre>

Covered by tests: omdurman-rules::src::effects::tests::cumulative_move_cost_may_not_exceed_allowance
omdurman-rules::src::effects::tests::wrong_faction_move_is_rejected
omdurman-rules::src::effects::tests::wrong_faction_gunboat_move_is_rejected

§5.13 – No MP accumulation between turns

implemented

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A unit may never accumulate movement points from turn to turn, nor may a unit transfer unused movement points to other units. A unit's unused movement points in any given turn are considered lost.

File	Symbol	Code Snippet
omdurman-rules/src/effects/dispatch.rs:298 GH:omdurman-rules/src/effects/dispatch.rs:298	end_player_turn	<pre>296 297 /// End the current player's turn: recover disrupted units, switch active player, advance turn index (rulebook §4). 298 pub fn end_player_turn(state: &mut GameState) { 299 #[cfg(not(feature = "kani"))] 300 debug!(</pre>

Covered by tests: omdurman-rules::src::effects::tests::recover_unit_is_the_owners_end_of_turn_only
omdurman-rules::src::effects::tests::unused_movement_points_do_not_carry_over

§5.21 – Friendlies transport via gunboat

implemented

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In general, naval transport missions are not allowed, i.e. gunboats may not carry any land units. The sole exception is that the Anglo-Egyptian player may transport the surviving units of the “Friendlies” brigade from the east bank of the Nile to the west bank after, and only after, the Dervish east bank unit (Isa Zachneih) has been eliminated. The transport is accomplished in the following sequence: a) on any turn that a “Friendlies” unit and any Anglo-Egyptian gunboat start their turn adjacent, that unit may load onto (i.e. stack with) the gunboat; b) during the Anglo-Egyptian player’s next turn the gunboat may move to any Nile hex adjacent to a west bank hex (up to the gunboat’s movement allowance); c) on the Anglo-Egyptian player’s third turn the “Friendlies” unit may disembark and move normally, paying the normal terrain cost for the first hex entered. The gunboat may also move normally that turn.

(see manual for full text)

File	Symbol	Code Snippet
omdurman-rules/src/unit.rs:317 GH:omdurman-rules/src/unit.rs:317	is_friendlys	<pre> 315 /// "Friendlies" units obey several special rules (\$5.21, \$5.23, \$6.52, 316 /// \$9.14 victory conditions). 317 pub fn is_friendlys(&self) -> bool { 318 matches!(319 self, </pre>
omdurman-rules/src/effects/state/movement.rs:487 GH:omdurman-rules/src/effects/state/movement.rs:487	friendlys_transport_offer	<pre> 485 /// regardless of selection. Pairs with [`GameEffect::FriendlysTransport`] 486 /// so the UI can offer exactly the action the engine would accept. 487 pub fn friendlys_transport_offer(&self, selected: Option<UnitId>) -> Option<FriendlysAction> { 488 match self.friendlys_transport { 489 None => { </pre>
omdurman-rules/src/unit.rs:522 GH:omdurman-rules/src/unit.rs:522	loaded_on	<pre> 520 pub disrupted: bool, 521 /// `Some(gunboat)` after a "Friendlies" unit loads onto a gunboat (\$5.21). 522 pub loaded_on: Option<UnitId>, 523 /// Set while the unit is building Zariba hexsides -- neither offensive 524 /// fire nor melee allowed that turn (\$5.3). </pre>
omdurman-rules/src/effects/effect.rs:197 GH:omdurman-rules/src/effects/effect.rs:197	FriendlysTransport	<pre> 195 196 /// Load/disembark the "Friendlies" brigade via gunboat (rulebook \$5.21). 197 FriendlysTransport(crate::FriendlysAction), 198 199 // -- Optional rules ----- </pre>
omdurman-rules/src/effects/river.rs:12 GH:omdurman-rules/src/effects/river.rs:12	apply_friendlys_transpo	<pre> 10 /// unit is freed (a disembarking `MoveUnit` should follow, costed by 11 /// terrain). 12 pub fn apply_friendlys_transport(13 state: &mut GameState, 14 action: FriendlysAction, </pre>
omdurman-rules/src/transport.rs:20 GH:omdurman-rules/src/transport.rs:20	FriendlysAction	<pre> 18 /// tracks each unit-gunboat pair independently. 19 #[derive(Serialize, Deserialize, Clone, Copy, PartialEq, Eq, Debug)] 20 pub enum FriendlysAction { 21 /// Turn N (the load turn): unit and gunboat started adjacent; unit 22 /// loads onto (stacks with) the gunboat. </pre>
omdurman-rules/src/transport.rs:40 GH:omdurman-rules/src/transport.rs:40	TransportState	<pre> 38 /// third turn. 39 #[derive(Serialize, Deserialize, Clone, Copy, PartialEq, Eq, Debug)] 40 pub enum TransportState { 41 /// Turn N (the load turn): unit and gunboat started adjacent; unit 42 /// loads onto (stacks with) the gunboat. </pre>

Covered by tests: [omdurman-rules::src::effects::tests::a_mined_gunboat_scores_and_takes_its_friendlys_down](#)
[omdurman-rules::src::effects::tests::friendlys_transport_offer_load_requires_prerequisites](#)
[omdurman-rules::src::effects::tests::friendlys_transport_offer_follows_state_machine](#)

§5.22 – Land units may never enter a Nile River hex

[implemented](#)

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With the exception of 5.21, land units may never enter a Nile River hex. Only gunboats may enter and move along Nile River hexes.

See also: §5.21

File	Symbol	Code Snippet
omdurman-rules/src/effects/state/movement.rs:43 GH:omdurman-rules/src/effects/state/movement.rs:43	can_move_unit_to	<pre>41 /// 42 /// Stacking (§5.51) is checked when the move is applied, not here. 43 pub fn can_move_unit_to(44 &self, 45 unit_id: UnitId,</pre>
omdurman-rules/src/effects/state/setup.rs:215 GH:omdurman-rules/src/effects/state/setup.rs:215	in_deployment_zone	<pre>213 /// plan / UI rather than this hex predicate. Documented, not silently 214 /// dropped. 215 pub fn in_deployment_zone(&self, player: Player, hex: HexCoord, is_boat: bool) -> bool { 216 // No board attached -> permissive (unit tests, unbound session). 217 if self.board.terrain.is_empty() {</pre>

Covered by tests: [omdurman-rules::src::effects::tests::campaign_deployment_is_boat_land_exclusive](#)
[omdurman-rules::src::effects::tests::fok_ae_gunboat_deploys_only_on_nile](#)
[omdurman-rules::src::effects::tests::fok_ae_land_unit_rejected_on_nile](#)
[omdurman-rules::src::effects::tests::deploy_via_real_sprite_resolution_matches_engine](#)
[omdurman-rules::src::effects::tests::fok_dervish_land_unit_rejected_on_nile](#)
[omdurman-rules::src::effects::tests::retreat_before_melee_may_not_land_on_nile](#)

§5.23 – Walled city entry restrictions

[implemented](#)

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Only certain units may enter the walled portion of Omdurman. For the Dervish player these are the Khalifa unit, the three Dervish artillery units, and the Taiasha units (the Khalifa's bodyguard). Any Anglo-Egyptian units that can get to the walled city may enter it (except gunboats and "Friendlys"). Units entering and/or exiting the walled city may only do so through a gate or breach hexside.

File	Symbol	Code Snippet
omdurman-types/src/lib.rs:152 GH:omdurman-types/src/lib.rs:152	HexsideRef	<pre>150 /// data by [`HexsideRef`]. 151 #[derive(Serialize, Deserialize, Clone, Copy, PartialEq, Eq, PartialOrd, Ord, Hash, Debug)] 152 pub struct HexsideRef { 153 pub a: HexCoord, 154 pub b: HexCoord,</pre>
omdurman-types/src/lib.rs:187 GH:omdurman-types/src/lib.rs:187	HexsideKind	<pre>185 strum::EnumIter, 186)] 187 pub enum HexsideKind { 188 /// City wall (Khartoum, walled city of Omdurman).</pre>

		Blocks LOS, blocks 189 /// movement except across gates/breaches (§5.23), blocks ZOC into the city
omdurman-types/src/lib.rs:258 GH:omdurman-types/src/lib.rs:258	blocks_movement	256 /// `omdurman-rules`). The trench *end* variants are therefore intentionally 257 /// not blocking. 258 pub fn blocks_movement(self) -> bool { 259 matches!(260 self,
omdurman-rules/src/unit.rs:343 GH:omdurman-rules/src/unit.rs:343	may_enter_walled_city	341 /// Taiasha bodyguard may enter. Anglo-Egyptian: any unit that can reach the 342 /// walled city *except* gunboats and "Friendlies". 343 pub fn may_enter_walled_city(&self) -> bool { 344 match self { 345 // §5.23 Dervish: Khalifa, artillery, Taiasha.
omdurman-rules/src/board.rs:250 GH:omdurman-rules/src/board.rs:250	is_walled_city	248 /// are its seeds). The set is derived once from the board data, replacing 249 /// the older "at least two of six hexsides are Wall/Gate/Breach" heuristic. 250 pub fn is_walled_city(&self, hex: HexCoord) -> bool { 251 // Membership in the precomputed enclosed area (see `walled_city`). 252 // Palace/Tomb hexes are always part of it (they are the seeds).
omdurman-rules/src/effects/error.rs:93 GH:omdurman-rules/src/effects/error.rs:93	WalledCityEntry	91 92 #[error("unit {0:?} is not eligible to enter the walled city of Omdurman at {1:?} (§5.23)")] 93 WalledCityEntry(UnitId, HexCoord), 94 95 #[error("movement cost {cost:?} exceeds allowance {allowance:?}")]

Proven

by:

[omdurman-types::src::lib::hexside_ref_is_order_independent](#) [omdurman-types::src::lib::hexside_blocking_classifiers_are_exact](#)

Covered

by

tests:

[omdurman-rules::src::effects::tests::a_wall_blocks_every_step_not_just_the_endpoints](#)
[omdurman-rules::src::effects::tests::can_move_rejects_wall_hexside](#) [omdurman-rules::src::effects::tests::can_move_allows_gate_hexside](#)
[omdurman-rules::src::effects::tests::walled_city_entry_allows_khalifa](#)
[omdurman-rules::src::effects::tests::walled_city_entry_rejects_unauthorized_dervish](#)
[omdurman-rules::src::effects::tests::walled_city_entry_rejects_ae_gunboat](#)
[omdurman-rules::src::effects::tests::walled_city_entry_not_enforced_for_fok](#)
[omdurman-rules::src::board_data::campaign_walled_city_is_enclosed_by_walls](#)
[omdurman-rules::src::effects::tests::breach_is_an_opening_for_los_and_effective_kinds](#)

§5.24 – Gunboat upstream/downstream movement

implemented

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Note that gunboats have two movement allowances separated by a slash, e.g. 10/16. The smaller number is the movement allowance when moving upstream, i.e. against the current (the direction of the current is indicated by arrows in the Nile). The larger number is the movement allowance when moving downstream, i.e. with the current. Gunboats may combine movement in both directions, but if they move even one hex upstream, their upstream movement allowance is their maximum movement allowance for that turn, and may not be exceeded.

File	Symbol	Code Snippet
omdurman-rules/src/unit.rs:243 GH:omdurman-rules/src/unit.rs:243	GunboatMovement	<pre> 241 /// the turn. 242 #[derive(Serialize, Deserialize, Clone, Copy, 243 pub struct GunboatMovement { 244 pub upstream: MovementAllowance, 245 pub downstream: MovementAllowance,</pre>
omdurman-types/src/lib.rs:967 GH:omdurman-types/src/lib.rs:967	is_boat	<pre> 965 966 /// Gunboats use the split upstream/downstream movement allowance (§5.24). 967 pub fn is_boat(self) -> bool { 968 matches!(self, UnitKind::Gunboat { .. }) 969 }</pre>

Proven by: [omdurman-types::src::lib::unit_kind_fire_and_movement_capability_law_is_exact](#)
[omdurman-types::src::lib::hex_direction_from_index_is_total_mod_six](#)
[omdurman-types::src::lib::nile_flow_rotation_composes_and_never_leaves_the_nile](#)

Covered by tests: [omdurman-rules::src::unit_profiles::boat_annotation_yields_split_gunboat_movement](#)
[omdurman-rules::src::effects::tests::gunboat_upstream_cap_is_sticky_across_moves](#)
[omdurman-app::src::picker::movement::gunboat_cap_gate_mirrors_the_engine](#)
[omdurman-app::src::picker::movement::remaining_mp_respects_the_sticky_upstream_cap](#)

§5.25 – Dervish forts may not move

implemented

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Dervish forts may not move in any way once placed.

File	Symbol	Code Snippet
omdurman-rules/src/unit.rs:506 GH:omdurman-rules/src/unit.rs:506	Immobile	<pre> 504 Gunboat(GunboatMovement), 505 /// Forts may not move once placed (§5.25). 506 Immobile, 507 } 508 </pre>
omdurman-types/src/lib.rs:924 GH:omdurman-types/src/lib.rs:924	UnitKind::Fort	<pre> 922 /// Permanent emplacement (§6.54): fire (artillery) / melee (defensive). 923 /// May not move once placed (§5.25). 924 Fort { fire: i32, melee: i32 }, 925 /// Dervish leader (§6.51): fire / melee / movement. May melee attack (§7.4). 926 DervishLeader {</pre>
omdurman-rules/src/unit.rs:506 GH:omdurman-rules/src/unit.rs:506	UnitMovement::Immobile	<pre> 504 Gunboat(GunboatMovement), 505 /// Forts may not move once placed (§5.25). 506 Immobile, 507 } 508 </pre>

Covered by tests: [omdurman-rules::src::effects::tests::forts_are_never_advance_eligible](#)
[omdurman-rules::src::effects::tests::immobile_fort_rejects_move_unit](#)

§5.26 – Units stop on entering enemy ZOC

implemented

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Units must stop their movement immediately upon entering an enemy zone of control (see 5.4).

See also: §5.4

File	Symbol	Code Snippet
omdurman-rules/src/effects/state/movement.rs:43 GH:omdurman-rules/src/effects/state/movement.rs:43	can_move_unit_to	<pre>41 /// 42 /// Stacking (§5.51) is checked when the move is applied, not here. 43 pub fn can_move_unit_to(44 &self, 45 unit_id: UnitId,</pre>
omdurman-rules/src/effects/state/stacking.rs:218 GH:omdurman-rules/src/effects/state/stacking.rs:218	hex_in_enemy_zoc	<pre>216 /// does not extend into or out of a Nile hex. With no board loaded these 217 /// reduce to the plain adjacency rule. 218 pub fn hex_in_enemy_zoc(219 &self, 220 hex: HexCoord,</pre>

Covered by tests: [omdurman-rules::src::effects::tests::unit_entering_enemy_zoc_may_move_no_further_that_turn](#)
[omdurman-rules::src::effects::tests::zoc_transit_check_uses_the_actual_path](#)

§5.41 – All units except AE leaders exert ZOC

implemented

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All units except Anglo-Egyptian leaders exert a zone of control (hereafter called a ZOC) into their six adjacent hexes (exception: Gunboats exert a ZOC only against enemy gunboats). Disrupted units have no ZOC.

File	Symbol	Code Snippet
omdurman-rules/src/effects/state/stacking.rs:110 GH:omdurman-rules/src/effects/state/stacking.rs:110	unit_projects_zoc_rule	<pre>108 /// * A fort projects ZOC out of its hex even when unoccupied (§5.44), 109 /// modelled by the fort unit projecting normally. 110 pub fn unit_projects_zoc_rule(111 unit: &UnitPlacement, 112 mover_player: Player,</pre>
omdurman-rules/src/unit.rs:561 GH:omdurman-rules/src/unit.rs:561	ZocReason	<pre>559 /// Used by the engine when answering "is this hex in an enemy ZOC?". 560 #[derive(Serialize, Deserialize, Clone, Copy, PartialEq, Eq, Debug)] 561 pub enum ZocReason { 562 /// Normal ZOC: any non-disrupted unit other than an Anglo-Egyptian 563 /// leader (§5.41) projects ZOC into each of its six adjacent hexes.</pre>
omdurman-rules/src/effects/state/stacking.rs:200 GH:omdurman-rules/src/effects/state/stacking.rs:200	unit_projects_zoc	<pre>198 /// §5.44) need the game map, which the engine does not hold; the app layers 199 /// those on top. This is the position/kind/ disruption core of the rule. 200 pub fn unit_projects_zoc(201 &self, 202 unit: &UnitPlacement,</pre>
omdurman-rules/src/ef-	zoc_hexes	<pre>255 /// ZOC covers a given hex; this function returns *which* hexes a</pre>

fects/state/ stacking.rs:257 GH:omdurman- rules/src/ef- fects/state/ stacking.rs:257		256 /// specific unit covers. 257 pub fn zoc_hexes{ 258 &self, 259 unit: &UnitPlacement,
---	--	--

Proven by: [omdurman-rules::src::effects::unit_projects_zoc_matches_manual_clauses](#)
[omdurman-types::src::lib::player_opponent_is_a_fixed_point_free_involution](#)

Covered by tests: [omdurman-rules::src::effects::tests::zoc_hexes_empty_for_anglo_egyptian_leader](#)
[omdurman-rules::src::effects::tests::zoc_hexes_normal_unit_projects_six_adjacent_minus_exclusions](#)
[omdurman-rules::src::effects::tests::zoc_hexes_empty_for_disrupted_unit](#)
[omdurman-rules::src::effects::tests::zoc_hexes_matches_hex_in_enemy_zoc](#) [omdurman-rules::src::effects::tests::zoc_hexes_excludes_nile](#)
[omdurman-rules::src::effects::tests::zoc_hexes_excludes_khor](#)

§5.42 – No MP cost to enter/leave enemy ZOC

implemented

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There is no movement point cost to enter or leave an enemy ZOC.

File	Symbol	Code Snippet
omdurman- rules/src/ef- fects/state/ movement.rs:432 GH:omdurman- rules/src/ef- fects/state/ movement.rs:432	movement_cost_for	430 /// 431 /// §5.42: entering or leaving an enemy ZOC adds no MP cost. 432 pub fn movement_cost_for{ 433 &self, 434 unit: &UnitPlacement,

Covered by tests: [omdurman-rules::src::effects::tests::entering_enemy_zoc_costs_no_extra_mp](#)

§5.43 – Units stop when entering enemy ZOC

implemented

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All units must stop when they enter an enemy ZOC and may move no further that turn. In their next movement phase they may withdraw or, if desired, move directly into another enemy ZOC.

File	Symbol	Code Snippet
omdurman- rules/src/ef- fects/state/ movement.rs:43 GH:omdurman- rules/src/ef- fects/state/ movement.rs:43	can_move_unit_to	41 /// 42 /// Stacking (§5.51) is checked when the move is applied, not here. 43 pub fn can_move_unit_to{ 44 &self, 45 unit_id: UnitId,
omdurman- rules/src/ef- fects/state/ stacking.rs:218 GH:omdurman- rules/src/ef- fects/state/ stacking.rs:218	hex_in_enemy_zoc	216 /// does not extend into or out of a Nile hex. With no board loaded these 217 /// reduce to the plain adjacency rule. 218 pub fn hex_in_enemy_zoc{ 219 &self, 220 hex: HexCoord,

§5.44 – ZOC limitations (walls, khor, fort, Nile, Zariba)

implemented

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ZOCs do not extend into or out of a Nile River hex (exception: Gunboats, see 5.41). ZOCs do not extend across a khor, into a fort, or into a hex inside the walled city across a wall hexside. ZOCs do extend out of a fort (even if unoccupied), and from a walled city hex into an adjacent non-walled-city hex across a wall hexside. ZOCs also extend out of (but not into) a walled city hex across a gate hexside. ZOCs extend both ways across a breach hexside. ZOCs also extend out of, but not into, a hut or building hex. In the historical scenario ZOCs extend out of, but not into, the Zariba across a Zariba hexside (also in the campaign game if the Zariba is constructed).

(see manual for full text)

See also: §5.41

File	Symbol	Code Snippet
omdurman-rules/src/unit.rs:561 GH:omdurman-rules/src/unit.rs:561	ZocReason	<pre> 559 /// Used by the engine when answering "is this hex in an enemy ZOC?". 560 #[derive(Serialize, Deserialize, Clone, Copy, PartialEq, Eq, Debug)] 561 pub enum ZocReason { 562 /// Normal ZOC: any non-disrupted unit other than an Anglo-Egyptian 563 /// leader (§5.41) projects ZOC into each of its six adjacent hexes.</pre>
omdurman-types/src/lib.rs:192 GH:omdurman-types/src/lib.rs:192	Wall	<pre> 190 /// (§5.44), blocks melee (§7.2), blocks advance- after-combat (§6.82). 191 #[default] 192 Wall, 193 /// Gate hexside in a wall. ZOC extends *out of* the walled city through 194 /// gates but not into it (§5.44). Melee may be made through a gate (§7.2).</pre>
omdurman-types/src/lib.rs:201 GH:omdurman-types/src/lib.rs:201	Khor	<pre> 199 /// Khor -- gully/wadi. ZOCs do not extend across (§5.44); advance after 200 /// combat may not cross (§6.82). 201 Khor, 202 /// Crest line. Blocks LOS unless the firer is on the higher side 203 /// (§6.3 note 7).</pre>
omdurman-rules/src/unit.rs:574 GH:omdurman-rules/src/unit.rs:574	ZocReason::Zariba	<pre> 572 /// Zariba hexside ZOC behaviour in the historical scenario / when the 573 /// Zariba is constructed (§5.44). 574 Zariba, 575 } 576 </pre>
omdurman-rules/src/effects/state/stacking.rs:200 GH:omdurman-rules/src/effects/state/stacking.rs:200	unit_projects_zoc	<pre> 198 /// §5.44) need the game map, which the engine does not hold; the app layers 199 /// those on top. This is the position/kind/ disruption core of the rule. 200 pub fn unit_projects_zoc(201 &self, 202 unit: &UnitPlacement,</pre>

omdurman-types/src/lib.rs:275 GH:omdurman-types/src/lib.rs:275	HexsideKind::blocks_zoc	<pre> 273 /// cannot express; those are left to the caller. 274 This predicate captures the 275 /// symmetric "does not extend across" cases. 276 pub fn blocks_zoc(self) -> bool { 277 matches!(278 self, </pre>
omdurman-rules/src/effects/state/stacking.rs:218 GH:omdurman-rules/src/effects/state/stacking.rs:218	hex_in_enemy_zoc	<pre> 216 /// does not extend into or out of a Nile hex. With 217 no board loaded these 218 /// reduce to the plain adjacency rule. 219 pub fn hex_in_enemy_zoc(220 &self, 221 hex: HexCoord, </pre>

Proven by: [omdurman-rules::src::effects::unit_projects_zoc_matches_manual_clauses](#)
[omdurman-types::src::lib::hexside_blocking_classifiers_are_exact](#)

Covered by tests: [omdurman-rules::src::effects::tests::zoc_hexes_excludes_nile](#) [omdurman-rules::src::effects::tests::zoc_hexes_excludes_khor](#)
[omdurman-rules::src::effects::tests::zoc_hexes_normal_unit_projects_six_adjacent_minus_exclusions](#)

§5.51 – Stacking limit (4 units + leaders, gunboats isolated)

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No more than four units may occupy a hex, with the exception of leaders and gunboats. All leader units are free stacking, i.e. they may stack in addition to the four-unit-per-hex stacking limitation. Gunboats may not stack with any other unit (Exception: 5.21). Players may move through friendly units at no additional cost in movement points. The stacking limitation applies only at the end of the movement phase and during combat.

See also: §5.21

File	Symbol	Code Snippet
omdurman-rules/src/unit.rs:586 GH:omdurman-rules/src/unit.rs:586	OverLimit	<pre> 584 /// (§6.51: a Dervish unit occupying his hex 585 eliminates him). 586 #[error("hex stack exceeds the four-unit limit 587 [\$5.51]")] 588 OverLimit, 589 /// "Gunboats may not stack with any other unit" 590 (§5.51, exception §5.21). 591 #[error("gunboats may not stack with non-gunboat 592 units [\$5.51]")] </pre>
omdurman-rules/src/unit.rs:589 GH:omdurman-rules/src/unit.rs:589	GunboatStack	<pre> 587 /// "Gunboats may not stack with any other unit" 588 (§5.51, exception §5.21). 589 #[error("gunboats may not stack with non-gunboat 590 units [\$5.51]")] 591 GunboatStack, 592 /// "Units of different Dervish tribes may not 593 stack together" (§5.52). 594 /// The Dervish artillery (§9.322's three guns) is 595 not a tribe but is not </pre>
omdurman-rules/src/unit.rs:599 GH:omdurman-rules/src/unit.rs:599	EnemyCohabitation	<pre> 597 /// lone Anglo-Egyptian leader is exempt (§6.51). 598 #[error("enemy units may not share a hex; melee, 599 not movement, engages them [\$5.51, §7.1]")] 600 EnemyCohabitation, 601 /// "If Dervish leaders elect to stack, they may 602 only stack with units of 603 their command (i.e. colour)" (§5.53). </pre>
omdurman-rules/src/effects/state/stacking.rs:218 GH:omdurman-rules/src/effects/state/stacking.rs:218	check_stacking	<pre> 149 /// Dervish artillery is its own group) may not 150 stack together. 151 /// * §5.53 -- a Dervish leader may stack only with </pre>

stacking.rs:151 GH:omdurman-rules/src/effects/state/stacking.rs:151		units of its command. 151 pub fn check_stacking(152 &self, 153 mover: &UnitPlacement,
omdurman-rules/src/effects/state/stacking.rs:21 GH:omdurman-rules/src/effects/state/stacking.rs:21	stacking_rule	19 /// artillery as its own group) may not share a hex. 20 /// * §5.53: a Dervish leader stacks only with units of its command. 21 pub fn stacking_rule(occupants: &[&UnitPlacement]) -> Result<(), crate::StackingError> { 22 use crate::StackingError; 23

Proven by: omdurman-rules::src::effects::stacking_rule_cohabitation_is_exact
omdurman-rules::src::effects::stacking_rule_is_symmetric omdurman-rules::src::effects::stacking_rule_leaders_are_free_stacking
omdurman-rules::src::effects::stacking_rule_limit_is_four_counted_units
omdurman-rules::src::effects::stacking_rule_gunboat_never_shares

Covered by tests: omdurman-rules::src::effects::tests::stacking_over_limit_rejected
omdurman-rules::src::effects::tests::mid_move_stacking_allows_pass_through
omdurman-rules::src::effects::tests::mid_move_stacking_rejects_over_limit_destination
omdurman-rules::src::effects::tests::validate_stacking_invariants_clean_state
omdurman-rules::src::effects::tests::validate_stacking_invariants_catches_stacking_violation
omdurman-rules::src::effects::tests::validate_stacking_invariants_allows_leaders_stacking
omdurman-rules::src::effects::tests::deploy_rejects_enemy_cohabitation_during_setup
omdurman-rules::src::effects::tests::validate_stacking_invariants_catches_enemy_cohabitation
omdurman-rules::src::effects::tests::ae_garrison_stacks_freely_under_gordon
omdurman-rules::src::effects::tests::melee_mandatory_advance_leaders_are_free_stacking

§5.52 – Different Dervish tribes may not stack together

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The units of different Dervish tribes may not stack together, even if they are the same color (e.g. although both are green, Mulazmin and Jehadia units may not stack with each other).

File	Symbol	Code Snippet
omdurman-rules/src/unit.rs:595 GH:omdurman-rules/src/unit.rs:595	DervishTribeMix	593 /// [`UnitIdentity::dervish_stacking_group`]). 594 #[error("Dervish units of different tribes may not stack [§5.52]")] 595 DervishTribeMix, 596 /// A unit may never share a hex with enemy units (§5.51, §7.1); only the 597 /// lone Anglo-Egyptian leader is exempt (§6.51).
omdurman-rules/src/unit.rs:395 GH:omdurman-rules/src/unit.rs:395	dervish_stacking_group	393 /// `None` for every unit the §5.52 law does not constrain: Dervish 394 /// leaders, forts and gunboats, and all Anglo-Egyptian units. 395 pub fn dervish_stacking_group(&self) -> Option<DervishStackingGroup> { 396 match self { 397 UnitIdentity::DervishTribal { tribe } => Some(DervishStackingGroup::Tribe(*tribe)),

Proven by: omdurman-rules::src::effects::stacking_rule_group_purity_is_exact

Covered by tests: omdurman-rules::src::unit_profiles::green_sections_are_mulazmin_tribal_units
omdurman-rules::src::effects::tests::deploy_rejects_dervish_tribe_mix
omdurman-rules::src::effects::tests::deploy_rejects_hadendowa_on_dervish_gun
omdurman-rules::src::effects::tests::dervish_guns_stack_with_guns_and_their_leader
omdurman-rules::src::effects::tests::validate_stacking_invariants_clean_state

§5.53 – Leader stacking with command colour only

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Leader units are not required to stack. If Dervish leaders elect to stack, however, they may only stack with units of their command (i.e. color). For example, Sheik El Din may only stack with Mulazmins or Jehadias.

****5.54) Anglo-Egyptian Brigade Integrity:**** All British, Sudanese, and Egyptian infantry units have their brigade designation printed in the upper right corner (e.g. “2B” = 2nd British Brigade; “3E” = 3rd Egyptian Brigade, etc.). In any combat phase in which all four infantry battalions belonging to any Anglo-Egyptian infantry brigade are stacked in the same hex they are said to have brigade integrity. Stacks having brigade integrity receive a +1 modifier to their fire combat die roll provided they all fire at the same enemy occupied hex. This modifier is in addition to the normal +1 bonus given to all Anglo-Egyptian direct fire attacks (see 6.24).

(see manual for full text)

See also: §5.54, §6.24

File	Symbol	Code Snippet
omdurman-rules/src/unit.rs:603 GH:omdurman-rules/src/unit.rs:603	DervishLeaderCommandMismatch	<pre>601 /// their command (i.e. colour)" (§5.53). 602 #[error("Dervish leader may only stack with units 603 of their own command [§5.53]")] 604 DervishLeaderCommandMismatch, 605 }</pre>

Covered by tests: [omdurman-rules::src::effects::tests::dervish_leader_stacks_only_with_command_colour](#)
[omdurman-rules::src::unit_profiles::command_owns_unit_follows_tribes_and_brigades](#)

§5.54 – Anglo-Egyptian Brigade Integrity

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File	Symbol	Code Snippet
omdurman-rules/src/unit.rs:609 GH:omdurman-rules/src/unit.rs:609	BrigadeIntegrity	<pre>607 /// stack contains all four battalions of a single 608 Anglo-Egyptian brigade. 609 #[derive(Serialize, Deserialize, Clone, Copy, 610 PartialEq, Eq, Debug)] 611 pub enum BrigadeIntegrity { 612 None, 613 Integrated(BrigadeId), 614 }</pre>
omdurman-rules/src/unit.rs:459 GH:omdurman-rules/src/unit.rs:459	brigade_integrity	<pre>457 /// Only a full stack of four battalions qualifies. 458 Three or fewer may still 459 stack and fire, but they receive no brigade- 460 integrity bonus. 461 pub fn brigade_integrity(identities: &[UnitIdentity]) - 462 > BrigadeIntegrity { 463 let Some(brigade) = identities.first().and_then(i 464 i.brigade()) else { 465 return BrigadeIntegrity::None; 466 } 467 Integrated(BrigadeId::new(brigade)) 468 }</pre>
omdurman-rules/src/unit.rs:609 GH:omdurman-rules/src/unit.rs:609	FireModifier::BrigadeIntegrity	<pre>607 /// stack contains all four battalions of a single 608 Anglo-Egyptian brigade. 609 #[derive(Serialize, Deserialize, Clone, Copy, 610 PartialEq, Eq, Debug)] 611 pub enum BrigadeIntegrity { 612 None, 613 Integrated(BrigadeId), 614 }</pre>
omdurman-rules/src/unit.rs:609 GH:omdurman-rules/src/unit.rs:609	BattalionOrdinal	<pre>18 /// brigade integrity requires all four stacked in 19 one hex (§5.54).</pre>

§6 – Fire Combat Phase

§6 – Fire Combat Phase

descriptive

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Fire Combat Phase

§6.1 – General Rules

descriptive

manual page unknown

General Rules

§6.2 – How To Have Fire Combat

descriptive

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How To Have Fire Combat

§6.3 – Line of Sight Table

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Line of Sight Table

This table is located on the back of this rulebook and should be self-explanatory. Locate the terrain type the firing unit is in and cross-index it with the terrain type the target unit is in. Terrain types in the intersecting box block line of sight, with exceptions as footnoted. Also study the “Special LOS Notes” given and remember that (with the exception of howitzer fire — see 6.64) you can’t fire at anything you can’t see!

See also: §6.64

File	Symbol	Code Snippet
omdurman-rules/src/los_table.rs:57 GH:omdurman-rules/src/los_table.rs:57	LosLevel	<pre>55 serde::Serialize, serde::Deserialize, Clone, Copy, 56 PartialEq, Eq, PartialOrd, Ord, Hash, Debug, 57)] 58 pub enum LosLevel { 59 Ground, 59 Rough,</pre>
omdurman-rules/src/los_table.rs:81 GH:omdurman-rules/src/los_table.rs:81	LosFeature	<pre>79 /// authored RON spellings. 80 #[derive(serde::Serialize, serde::Deserialize, Clone, 81 Copy, PartialEq, Eq, Hash, Debug)] 82 pub enum LosFeature { 83 /// A hex containing units (gunboats/forts excluded 83 Units,</pre>
omdurman-rules/src/los_table.rs:103 GH:omdurman-rules/src/los_table.rs:103	LosCondition	<pre>101 /// A positional condition from the LOS table Detail 102 footnotes. 103 #[derive(serde::Serialize, serde::Deserialize, Clone, 104 Copy, PartialEq, Eq, Hash, Debug)] 105 pub enum LosCondition { 106 /// (1) Blocks only if the ray passes through more</pre>

		than two such features. 105 MoreThanTwo,
omdurman-rules/src/los_table.rs:183 GH:omdurman-rules/src/los_table.rs:183	los_level	181 /// 182 /// For all other units, the level is derived from the terrain at `hex`. 183 pub fn los_level_for_unit(184 kind: UnitKind, 185 hex: HexCoord,
omdurman-rules/src/los_table.rs:183 GH:omdurman-rules/src/los_table.rs:183	los_level_for_unit	181 /// 182 /// For all other units, the level is derived from the terrain at `hex`. 183 pub fn los_level_for_unit(184 kind: UnitKind, 185 hex: HexCoord,
omdurman-rules/src/los_table.rs:223 GH:omdurman-rules/src/los_table.rs:223	blocking_rules	221 /// always blocks. Indexing is in-bounds by construction (both enums have 222 /// exactly three variants). 223 pub fn blocking_rules(firer: LosLevel, target: LosLevel) -> &'static [BlockingRule] { 224 crate::tables_data::LOS_CELLS[firer.index()][target.index()] 225 }
omdurman-rules/src/los_table.rs:558 GH:omdurman-rules/src/los_table.rs:558	has_los	556 /// board-only tools). 557 #[allow(clippy::too_many_arguments)] 558 pub fn has_los(559 board: &crate::board::BoardInfo, 560 from: HexCoord,
omdurman-types/src/lib.rs:230 GH:omdurman-types/src/lib.rs:230	HexsideKind::blocks_los	228 /// (LOS table conditions 2-4, 7) and note (e) are handled by the engine 229 /// in `omdurman_rules::los_table`, not by this predicate. 230 pub fn blocks_los(self) -> bool { 231 matches!(self, HexsideKind::Wall 232 HexsideKind::Crest)
omdurman-types/src/lib.rs:230 GH:omdurman-types/src/lib.rs:230	Terrain::blocks_los	228 /// (LOS table conditions 2-4, 7) and note (e) are handled by the engine 229 /// in `omdurman_rules::los_table`, not by this predicate. 230 pub fn blocks_los(self) -> bool { 231 matches!(self, HexsideKind::Wall 232 HexsideKind::Crest)
omdurman-types/src/lib.rs:452 GH:omdurman-types/src/lib.rs:452	Terrain::is_los_trees	450 /// (§6.3 note 1). Retained for compatibility; the full LOS engine 451 /// checks `Terrain::Trees` directly. 452 pub fn is_los_trees(self) -> bool { 453 matches!(self, Terrain::Trees { .. }) 454 }

Proven

by: [omdurman-types::src::lib::line_between_forms_a_connected_ray](#) [omdurman-types::src::lib::hexside_blocking_classifiers_are_exact](#)
[omdurman-rules::src::los_table::los_level_depends_only_on_the_ground](#)
[omdurman-rules::src::los_table::los_level_overrides_hold_for_gunboats_and_forts](#)
[omdurman-rules::src::los_table::blocking_grid_is_total_and_walls_block_ground_firers](#)

Covered by tests: [omdurman-rules::src::los_table::los_level_mapping](#) [omdurman-rules::src::los_table::blocking_rules_all_cells_covered](#)
[omdurman-rules::src::los_table::has_los_empty_board_is_clear](#) [omdurman-rules::src::los_table::has_los_adjacent_clear](#)
[omdurman-rules::src::los_table::has_los_howitzzer_bypasses](#)
[omdurman-rules::src::los_table::has_los_wall_hexside_blocks_ground_to_ground](#)
[omdurman-rules::src::los_table::has_los_gate_hexside_passes](#) [omdurman-rules::src::los_table::has_los_breach_hexside_passes](#)
[omdurman-rules::src::effects::tests::breach_is_an_opening_for_los_and_effective_kinds](#)
[omdurman-rules::src::los_table::has_los_rough_intervening_blocks_ground_to_ground](#)

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omdurman-rules::src::los_table::has_los_two_tree_hexes_pass_ground_to_ground
omdurman-rules::src::los_table::has_los_three_tree_hexes_block_ground_to_ground
omdurman-rules::src::los_table::has_los_two_hut_hexes_pass_ground_to_ground
omdurman-rules::src::los_table::has_los_three_hut_hexes_block_ground_to_ground
omdurman-rules::src::los_table::has_los_hilltop_to_hilltop_clear_no_units
omdurman-rules::src::los_table::has_los_hilltop_to_hilltop_blocked_by_hilltop_unit
omdurman-rules::src::los_table::has_los_hilltop_to_hilltop_not_blocked_by_ground_unit
omdurman-rules::src::los_table::has_los_rough_to_rough_unit_at_lower_level_passes
omdurman-rules::src::los_table::has_los_rough_to_rough_unit_at_same_level_blocks
omdurman-rules::src::los_table::has_los_rough_to_rough_hilltop_blocks
omdurman-rules::src::los_table::has_los_ground_to_hilltop_intervening_hilltop_blocks
omdurman-rules::src::los_table::has_los_building_blocks_like_huts_ground_to_ground
omdurman-rules::src::los_table::has_los_two_building_hexes_pass_ground_to_ground
omdurman-rules::src::los_table::los_level_for_unit_gunboat_is_rough
omdurman-rules::src::los_table::los_level_for_unit_fort_is_ground
omdurman-rules::src::los_table::los_level_for_unit_walled_city_adj_wall_is_rough
omdurman-rules::src::los_table::gunboat_firer_uses_rough_row_not_ground
omdurman-rules::src::los_table::los_reflexive_all_terrains
omdurman-rules::src::los_table::los_reflexive_hilltop
omdurman-rules::src::los_table::los_symmetric_ground_to_ground_no_units
omdurman-rules::src::los_table::los_howitzer_always_has_los
omdurman-rules::src::los_table::los_howitzer_same_hex
omdurman-rules::src::los_table::los_blocking_rules_match_reference_table
omdurman-rules::src::los_table::los_ground_to_ground_features_block_as_expected

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§6.4 – Fire Combat Sequence

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Fire Combat Sequence

The sequence of fire combat resolution is the same for both defensive and offensive fire combat. During the Dervish player turn the Anglo-Egyptian player executes 6.41 AND 6.42 as defensive fire, after which the Dervish player executes 6.41 as offensive fire. During the Anglo-Egyptian player turn the Dervish player executes 6.41 as defensive fire, after which the Anglo-Egyptian player executes 6.41 AND 6.42 as offensive fire.

****6.41) Direct Fire Subphase (Dervish and Anglo-Egyptian players):**** The firing player must first allocate all of his fire attacks, combining his units' direct fire combat factors in any manner he wishes. After all fire has been allocated, the firing player then resolves his attacks in any order he wishes.

****6.42) Maxim Second Fire and Howitzer Fire Subphase (Anglo-Egyptian player only):**** Anglo-Egyptian named gunboats may now fire their artillery factor as howitzer fire (see 6.64) and all Maxim guns may fire a second time. Once

(see manual for full text)

See also: §6.41, §6.42, §6.64

File	Symbol	Code Snippet
omdurman-rules/src/turn.rs:44 GH:omdurman-rules/src/turn.rs:44	FireSubPhase	<pre> 42 43 #[derive(Serialize, Deserialize, Clone, Copy, PartialEq, Eq, Hash, Debug)] 44 pub enum FireSubPhase { 45 /// Direct fire (§6.41). Both sides participate in this sub-phase. 46 DirectFire, </pre>
omdurman-rules/src/effects/effect.rs:73 GH:omdurman-rules/src/effects/effect.rs:73	FireCombat	<pre> 71 /// - Firers marked as fired; target hex marked as fired-at. 72 /// - Victory points awarded for eliminations. 73 FireCombat { attack: FireAttack, roll: DieRoll }, 74 75 /// Resolve a howitzer bombardment (two rolls: CRT + impact scatter) </pre>

Covered by tests: omdurman-rules::src::effects::tests::turn_advances_through_phases

§6.5 – Special Unit Capabilities

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Special Unit Capabilities

****6.51) Leader Units:**** Dervish leader units have a fire factor, a melee factor, and a movement factor. They may thus attack, melee, and be eliminated like any other combat unit. Their special benefit is that they stack free.

Anglo-Egyptian leaders have a movement factor only. They are eliminated if a) they are alone in a hex when a Dervish unit occupies or passes through that hex, or b) if all of the combat units a leader is stacked with are eliminated in fire combat or melee. The special function of Anglo-Egyptian leaders is that at least one must survive to occupy the Mahdi’s tomb hex if it is to be taken from the Dervish player (see 9.14).

****6.52) Anglo-Egyptian “Friendlies” Brigade:**** These units represent native volunteers in the Anglo-Egyptian army. They fire rifles on the Dervish Range Effects Table and melee with the Dervish melee modifier. They may not enter the walled city of Omdurman (see 5.23). They may be transferred to the west bank (see 5.21).

(see manual for full text)

See also: §5.21, §5.23, §5.44, §6.51, §6.52, §6.53, §6.54, §6.62, §6.63, §7.6, §9.14

§6.6 – Special Artillery Capabilities

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Special Artillery Capabilities

§6.7 – Defensive Fire

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Defensive Fire

In Defensive Fire phase, all of the non-moving player’s units may fire at any of the moving player’s units in range, within the limitations imposed by the rules of combat (see 6.1 to 6.6). There is no advance after combat as a result of defensive fires.

See also: §6.1, §6.6

File	Symbol	Code Snippet
omdurman-rules/src/effects/state/melee.rs:141 GH:omdurman-rules/src/effects/state/melee.rs:141	can_advance_after_comb	139 // player's unit, not artillery, adjacent to `to`, and `to` now empty. 140 // Wall/khor hexside restrictions are not enforced (no hexside map data). 141 pub fn can_advance_after_combat(&self, unit_id: UnitId, to: HexCoord) -> Result<(), RuleError> { 142 let unit = self.unit_or_err(unit_id)?; 143 // §6.7: there is no advance after combat as a result of defensive fire.

Covered by tests: omdurman-rules::src::effects::tests::no_advance_after_defensive_fire
omdurman-rules::src::effects::tests::defensive_fire_opens_no_advance_window

§6.8 – Offensive Fire

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Offensive Fire

§6.11 – Fire combat factor printed on units

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The fire combat factor of the various unit types is printed directly on the units and is a numerical expression of the unit's fire strength.

File	Symbol	Code Snippet
omdurman-rules/src/scalars.rs:18 GH:omdurman-rules/src/scalars.rs:18	FireFactor	<pre>16 // Every possible value from the annotated counter 17 set is a named variant. 18 #[derive(Serialize, Deserialize, Clone, Copy, 19 PartialEq, Eq, Hash, Debug, strum::Display)] 20 pub enum FireFactor { One = 1, Three = 3,</pre>

Covered by tests: [omdurman-rules::src::tests::fire_factor_sum_to_row](#)

§6.12 – Fire combat is always voluntary

implicit

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Fire combat is always voluntary. A unit is never required to fire at enemy units merely because they are in range or adjacent.

§6.13 – Fire factor is unitary (may not be divided)

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If a unit elects to fire, its fire combat factor at an enemy unit, that fire combat factor is unitary. A unit's fire combat factor may not be divided up to fire at enemy units on different hexes.

File	Symbol	Code Snippet
omdurman-rules/src/effects/fire.rs:502 GH:omdurman-rules/src/effects/fire.rs:502	build_fire_attack_from	<pre>500 // derived exactly as in [`build_fire_attack`]. 501 Returns `None` for an empty, 502 duplicated, or not-fully-legal list. 503 pub fn build_fire_attack_from(504 gs: &GameState, 505 firer_hex: HexCoord,</pre>

Covered by tests: [omdurman-rules::src::effects::tests::fire_rejects_duplicate_firers_and_a_foreign_firing_player](#)
[omdurman-rules::src::effects::tests::single_unit_fires_alone_from_a_stack](#)

§6.14 – Players may combine fire factors into one attack

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Players may combine fire during fire combat phase, i.e. they may fire at an enemy-occupied hex with as many friendly units as may legally do so, combining all of their fire combat factors into one attack. Note that in any given fire combat phase, however, a combat unit may only fire once and may only be fired at once (exceptions: Maxim guns and gunboats — see 6.4).

See also: §6.4

File	Symbol	Code Snippet
omdurman-rules/src/scalars.rs:32 GH:omdurman-rules/src/scalars.rs:32	sum_to_row	<pre> 30 impl FireFactor { 31 /// Sum multiple fire factors and return the corresponding Combat Results Table row (rulebook §6.11). 32 pub fn sum_to_row<'a>(factors: impl IntoIterator<Item = &'a FireFactor>) -> FireFactorRow { 33 let total: u16 = factors.into_iter().map(f f.value()).sum(); 34 crate::combat_results_table::FireFactorRow::from_total(total) </pre>
omdurman-rules/src/effects/state/fire.rs:49 GH:omdurman-rules/src/effects/state/fire.rs:49	GameS- tate::can_fire_at	<pre> 47 /// modifier in the [<code>FireAttack</code>] and is responsible for the LOS gate. 48 /// (Howitzer fire ignores LOS entirely -- §6.64.) 49 pub fn can_fire_at(50 &self, 51 firer: UnitId, </pre>

Covered by tests: [omdurman-rules::src::effects::tests::dead_units_are_pruned_from_every_tracker](#)
[omdurman-rules::src::effects::tests::unit_may_only_be_fired_at_once_per_phase](#)
[omdurman-rules::src::effects::tests::gunboat_and_maxim_may_be_fired_at_repeatedly](#)
[omdurman-rules::src::effects::tests::rejected_fire_attack_does_not_mark_firers_as_fired](#)
[omdurman-rules::src::effects::tests::single_unit_fires_alone_from_a_stack](#)
[omdurman-rules::src::effects::tests::infantry_may_fire_only_once_per_subphase](#)
[omdurman-rules::src::effects::tests::maxim_may_fire_twice_per_turn](#)

§6.15 – May divide a stack to fire at different hexes

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Players may also divide a stack of units in order to fire at different enemy-occupied hexes. Anglo-Egyptian infantry units having brigade integrity, however, do not receive their +1 direct fire modifier unless they all fire at the same enemy-occupied hex (see 5.54).

See also: §5.54

File	Symbol	Code Snippet
omdurman-rules/src/effects/fire.rs:502 GH:omdurman-rules/src/effects/fire.rs:502	build_fire_attack_from	<pre> 500 /// derived exactly as in [<code>build_fire_attack</code>]. Returns <code>None</code> for an empty, 501 /// duplicated, or not-fully-legal list. 502 pub fn build_fire_attack_from(503 gs: &GameState, 504 firer_hex: HexCoord, </pre>

Covered by tests: [omdurman-rules::src::effects::tests::stack_may_split_its_fire_across_hexes](#)

§6.16 – Halving fire strength rounds down, minimum 1

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When halving fire combat strength, always round down each individual unit. For example, an Egyptian brigade of four battalions, each having a printed strength of 9 fire factors, will fire a total of 16 factors when halved. However, a unit's firing strength is never reduced below one by halving.

File	Symbol	Code Snippet
omdurman-rules/src/unit.rs:210	RangeBand	<pre> 208 /// multiplied at a given distance (§6.22). 209 #[derive(Serialize, Deserialize, Clone, Copy, PartialEq, Eq, Debug)] </pre>

GH:omdurman-rules/src/unit.rs:210		210 pub enum RangeBand { 211 Tripled, 212 Doubled,
-----------------------------------	--	--

Proven by: [omdurman-rules::src::verification::range_band_halved_is_max_of_one_and_floor_half](#)
[omdurman-rules::src::verification::range_band_apply_is_monotone_in_raw](#)
[omdurman-rules::src::verification::disrupt_half_is_rounded_up](#)

Covered by tests: [omdurman-rules::src::tests::halving_rounds_down_and_never_below_one](#)

§6.21 – First check LOS before firing

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When combat units wish to fire at enemy units, first check the Line of Sight Table to be sure the firing unit can see the target hex (exception: howitzer fire, see 6.64).

See also: §6.64

File	Symbol	Code Snippet
omdurman-rules/src/los_table.rs:223 GH:omdurman-rules/src/los_table.rs:223	blocking_rules	221 /// always blocks. Indexing is in-bounds by construction (both enums have 222 /// exactly three variants). 223 pub fn blocking_rules(firer: LosLevel, target: LosLevel) -> &'static [BlockingRule] { 224 crate::tables_data::LOS_CELLS[firer.index()] [target.index()] 225 }
omdurman-rules/src/los_table.rs:558 GH:omdurman-rules/src/los_table.rs:558	has_los	556 /// board-only tools). 557 #[allow(clippy::too_many_arguments)] 558 pub fn has_los(559 board: &crate::board::BoardInfo, 560 from: HexCoord,

Covered by [omdurman-rules::src::effects::tests::can_fire_at_rejects_blocked_los](#) [omdurman-rules::src::effects::tests::can_fire_at_allows_clear_los](#) tests:

§6.22 – Consult Range Effects Table

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Next consult the Range Effects Table to see if the firing unit's fire combat factor is tripled, doubled, normal, halved, or if the target hex is out of range. Add up the total number of fire combat factors firing at the enemy-occupied hex.

File	Symbol	Code Snippet
omdurman-rules/src/range_effects.rs:34 GH:omdurman-rules/src/range_effects.rs:34	ae_range_effects	32 /// Look up the range band for an Anglo-Egyptian weapon (\$6.22, \$6.24). 33 /// Distances > 10 are out of range for all weapons. 34 pub fn ae_range_effects(weapon: WeaponClass, distance: HexDistance) -> RangeBand { 35 band_at(faction_rows(true), weapon, distance) 36 }
omdurman-rules/src/range_effects.rs:40 GH:omdurman-rules/src/range_effects.rs:40	dervish_range_effects	38 /// Look up the range band for a Dervish weapon (\$6.22). 39 /// Distances > 10 are out of range for all weapons. 40 pub fn dervish_range_effects(weapon: WeaponClass, distance: HexDistance) -> RangeBand {

		<pre> 41 band_at(faction_rows(false), weapon, distance) 42 }</pre>
omdurman-rules/src/unit.rs:210 GH:omdurman-rules/src/unit.rs:210	RangeBand	<pre> 208 /// multiplied at a given distance (§6.22). 209 #[derive(Serialize, Deserialize, Clone, Copy, PartialEq, Eq, Debug)] 210 pub enum RangeBand { 211 Tripled, 212 Doubled,</pre>
omdurman-rules/src/scalars.rs:119 GH:omdurman-rules/src/scalars.rs:119	HexDistance	<pre> 117 /// (rulebook §6.22, §7.5). 118 #[derive(Serialize, Deserialize, Clone, Copy, PartialEq, Eq, PartialOrd, Ord, Hash, Debug)] 119 pub struct HexDistance(pub(crate) u16); 120 121 impl HexDistance {</pre>

Proven by: [omdurman-types::src::lib::adjacency_iff_distance_one](#)
[omdurman-rules::src::verification::fire_factor_row_from_total_matches_printed_bands](#)
[omdurman-rules::src::verification::range_band_multiplier_arithmetic_is_exact](#)
[omdurman-rules::src::range_effects::range_effects_are_out_of_range_outside_the_printed_ten_hex_distance_window](#)
[omdurman-rules::src::range_effects::max_day_range_is_the_last_in_range_hex](#)
[omdurman-rules::src::effects::fire::range_band_for_routes_to_the_right_faction_table](#)

Covered by tests:

[omdurman-rules::src::range_effects::ae_rifles_doubled_at_range_1](#)
[omdurman-rules::src::range_effects::ae_rifles_halved_at_range_4](#)
[omdurman-rules::src::range_effects::ae_howitzzer_range](#)
[omdurman-rules::src::range_effects::dervish_rifles_shorter_range](#)
[omdurman-rules::src::range_effects::melee_only_range_1](#)
[omdurman-rules::src::range_effects::ae_range_effects_artillery_full](#)
[omdurman-rules::src::range_effects::ae_range_effects_maxims_match_rifles](#)
[omdurman-rules::src::range_effects::ae_range_effects_distance_over_10](#)
[omdurman-rules::src::range_effects::ae_range_effects_howitzzer_halved_4_to_10](#)
[omdurman-rules::src::range_effects::dervish_range_effects_rifles](#)
[omdurman-rules::src::range_effects::dervish_range_effects_artillery](#)
[omdurman-rules::src::range_effects::dervish_range_effects_maxims_and_howitzzer](#)
[omdurman-rules::src::range_effects::dervish_range_effects_melee](#)
[omdurman-rules::src::range_effects::dervish_range_effects_distance_over_10](#)
[omdurman-rules::src::effects::tests::fire_combat_eliminate_target](#)
[omdurman-rules::src::range_effects::max_day_range_all_combos](#)
[omdurman-rules::src::range_effects::range_effects_every_cell_ae_artillery](#)
[omdurman-rules::src::range_effects::range_effects_every_cell_ae_howitzzer](#)
[omdurman-rules::src::range_effects::range_effects_every_cell_ae_maxims](#)
[omdurman-rules::src::range_effects::range_effects_every_cell_ae_rifles](#)
[omdurman-rules::src::range_effects::range_effects_every_cell_dervish_artillery](#)
[omdurman-rules::src::range_effects::range_effects_every_cell_dervish_maxims_howitzzer](#)
[omdurman-rules::src::range_effects::range_effects_every_cell_dervish_rifles](#)
[omdurman-rules::src::range_effects::range_effects_every_cell_dervish_spears](#)
[omdurman-rules::src::effects::tests::can_fire_at_gates_phase_range_and_player](#)
[omdurman-rules::src::effects::tests::mixed_attack_bands_per_firer](#)
[omdurman-rules::src::range_effects::ae_range_effects_monotone_non_increasing](#)
[omdurman-rules::src::range_effects::ae_howitzzer_has_minimum_range](#)
[omdurman-rules::src::range_effects::dervish_range_effects_monotone_non_increasing](#)
[omdurman-rules::src::range_effects::range_effects_first_range_max_effect_last_range_oor](#)

§6.23 – Terrain defensive modifier

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Next check the Terrain Effects Chart to see if the enemy-occupied hex fired upon contains any terrain which gives the enemy units in that hex a defensive benefit. If so, apply this negative modifier to the roll of the ten-sided die and cross-index your net die roll on the Combat Results Table with the number of combat factors firing.

File	Symbol	Code Snippet
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omdurman-rules/src/terrain_chart.rs:59 GH:omdurman-rules/src/terrain_chart.rs:59	defense_modifier	<pre> 57 58 /// Convenience: get the defense modifier for a terrain type (rulebook §6.23, Terrain Effects Chart). 59 pub fn defense_modifier(terrain: Terrain) -> i16 { 60 terrain_effects_chart(terrain).defense_modifier 61 } </pre>
omdurman-rules/src/combat.rs:26 GH:omdurman-rules/src/combat.rs:26	FireModifier::Terrain	<pre> 24 /// Negative modifier from the Terrain Effects Chart applied to the 25 /// defender's hex (§6.23). 26 Terrain(i16), 27 /// -2 thorn-hedge defensive modifier (§9.231). 28 ZaribaThornHedge, </pre>

Proven by: [omdurman-rules::src::verification::fire_modifier_keeps_roll_legal](#)
[omdurman-rules::src::terrain_chart::defence_column_never_helps_the_attacker](#)

Covered by tests: [omdurman-rules::src::terrain_chart::clear_terrain_no_bonus](#) [omdurman-rules::src::terrain_chart::building_gives_minus_3](#)
[omdurman-rules::src::terrain_chart::trees_cost_one_and_give_no_defense](#)
[omdurman-rules::src::terrain_chart::rough_movement_and_defense](#)
[omdurman-rules::src::terrain_chart::swamp_movement_and_defense](#)
[omdurman-rules::src::terrain_chart::hilltop_movement_and_defense](#)
[omdurman-rules::src::terrain_chart::huts_movement_and_defense](#)
[omdurman-rules::src::terrain_chart::defense_modifier_convenience_matches_chart](#)
[omdurman-rules::src::terrain_chart::terrain_movement_costs_in_bounds](#)
[omdurman-rules::src::terrain_chart::terrain_defense_modifier_non_positive](#)

§6.24 – Anglo-Egyptian direct fire accuracy bonus and brigade integrity

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All Anglo-Egyptian direct fire attacks receive a +1 modifier to their die roll as an accuracy bonus. In addition, any stack of Anglo-Egyptian infantry having brigade integrity (see 5.54) receives a +1 modifier to their die roll if all four fire at the same enemy-occupied hex. These modifiers are cumulative.

See also: §5.54

File	Symbol	Code Snippet
omdurman-rules/src/combat.rs:20 GH:omdurman-rules/src/combat.rs:20	AngloEgyptianDirectFire	<pre> 18 pub enum FireModifier { 19 /// +1 to all Anglo-Egyptian *direct* fire (§6.24). 20 AngloEgyptianDirectFire, 21 /// +1 brigade integrity, applied only if all four battalions fire at 22 /// the same enemy-occupied hex (§5.54, §6.24). </pre>
omdurman-rules/src/unit.rs:609 GH:omdurman-rules/src/unit.rs:609	BrigadeIntegrity	<pre> 607 /// stack contains all four battalions of a single Anglo-Egyptian brigade. 608 #[derive(Serialize, Deserialize, Clone, Copy, PartialEq, Eq, Debug)] 609 pub enum BrigadeIntegrity { 610 None, 611 Integrated(BrigadeId), </pre>
omdurman-rules/src/combat.rs:36 GH:omdurman-rules/src/combat.rs:36	FireModifier::die_modifier	<pre> 34 impl FireModifier { 35 /// Return the numeric die-roll modifier for this bonus/penalty (rulebook §6.24, §5.54, §6.23, §9.231, §9.232). 36 pub fn die_modifier(self) -> i16 { 37 match self { 38 FireModifier::AngloEgyptianDirectFire FireModifier::BrigadeIntegrity => 1, </pre>

Proven by: [omdurman-rules::src::verification::die_roll_apply_modifier_is_total](#)

Covered by tests: [omdurman-rules::src::effects::tests::brigade_integrity_modifier_is_engine_derived](#)
[omdurman-rules::src::effects::tests::fire_modifiers_are_engine_derived_and_mismatches_rejected](#)

§6.41 – Direct Fire Subphase

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File	Symbol	Code Snippet
omdurman-rules/src/turn.rs:46 GH:omdurman-rules/src/turn.rs:46	DirectFire	<pre>44 pub enum FireSubPhase { 45 /// Direct fire (§6.41). Both sides participate in this sub-phase. 46 DirectFire, 47 /// Anglo-Egyptian only: Maxim second fire + named- gunboat howitzer fire (§6.42). 48 MaximSecondAndHowitzer,</pre>

Covered by tests: [omdurman-rules::src::effects::tests::fire_rejects_duplicate_firers_and_a_foreign_firing_player](#)
[omdurman-rules::src::effects::tests::disrupted_unit_cannot_fire](#) [omdurman-rules::src::effects::tests::fire_combat_elimimates_target](#)

§6.42 – Maxim Second Fire and Howitzer Fire Subphase

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File	Symbol	Code Snippet
omdurman-rules/src/turn.rs:48 GH:omdurman-rules/src/turn.rs:48	MaximSecondAndHowitzer	<pre>46 DirectFire, 47 /// Anglo-Egyptian only: Maxim second fire + named- gunboat howitzer fire (§6.42). 48 MaximSecondAndHowitzer, 49 } 50 </pre>
omdurman-types/src/lib.rs:985 GH:omdurman-types/src/lib.rs:985	fires_twice	<pre>983 /// Maxim guns fire twice per turn -- once in the Direct Fire Subphase and 984 /// again in the Maxim Second Fire Subphase (rulebook §6.42). 985 pub fn fires_twice(self) -> bool { 986 matches!(self, UnitKind::Maxim { .. }) 987 }</pre>

Proven by: [omdurman-types::src::lib::unit_kind_fire_and_movement_capability_law_is_exact](#)

Covered by tests: [omdurman-rules::src::howitzer_scatter::howitzer_on_target_7_to_10](#)
[omdurman-rules::src::howitzer_scatter::howitzer_scatters_below_7](#)
[omdurman-rules::src::effects::tests::advance_window_bridges_fire_subphase_and_closes_at_melee](#)
[omdurman-rules::src::effects::tests::fired_at_tracker_resets_at_maxim_subphase](#)
[omdurman-rules::src::effects::tests::maxim_may_fire_twice_per_turn](#)
[omdurman-rules::src::effects::tests::maxim_that_skipped_direct_may_fire_once_in_second_subphase](#)

§6.51 – Leader Units

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File	Symbol	Code Snippet
omdurman-rules/src/unit.rs:108 GH:omdurman-rules/src/unit.rs:108	BritishLeader	<pre>106 /// to claim the Mahdi's Tomb (§9.14). 107 #[derive(Serialize, Deserialize, Clone, Copy, PartialEq, Eq, Hash, Debug, strum::Display)] 108 pub enum BritishLeader { 109 Kitchener, 110 Gatacre,</pre>

omdurman-types/src/lib.rs:932 GH:omdurman-types/src/lib.rs:932	BritishLeader	<pre> 930 }, 931 // Anglo-Egyptian leader (§6.51): movement only. 932 BritishLeader { movement: i32 }, 933 // Wall-breach marker placed by artillery fire (§6.63). Not a combat unit. 934 Breach,</pre>
omdurman-types/src/lib.rs:973 GH:omdurman-types/src/lib.rs:973	has_combat_factors	<pre> 971 // British leaders print a movement factor only (§6.51); other kinds carry 972 // fire and/or melee factors. Markers carry no stats. 973 pub fn has_combat_factors(self) -> bool { 974 !matches!(975 self,</pre>

Proven by: [omdurman-types::src::lib::unit_kind_fire_and_movement_capability_law_is_exact](#)

Covered by tests:

- [omdurman-rules::src::unit_profiles::zero_factor_is_none_not_zero](#)
- [omdurman-rules::src::unit_profiles::kitchener_block_resolves_leaders_friendlys_camel_and_sudanese](#)
- [omdurman-rules::src::unit_profiles::dervish_leader_sections_resolve_leader_and_retinue_per_cell](#)
- [omdurman-rules::src::effects::tests::deploy_rejects_enemy_cohabitation_during_setup](#)
- [omdurman-rules::src::effects::tests::dervish_move_through_lone_ae_leader_hex_eliminate_the_leader](#)
- [omdurman-rules::src::effects::tests::dervish_move_onto_lone_ae_leader_hex_eliminate_the_leader](#)
- [omdurman-rules::src::effects::tests::ae_leader_with_combat_unit_is_not_overrun_by_dervish_move](#)
- [omdurman-rules::src::effects::tests::ae_leader_eliminated_with_last_combat_unit_in_fire_combat](#)

§6.52 – Anglo-Egyptian Friendlys Brigade

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File	Symbol	Code Snippet
omdurman-rules/src/unit.rs:317 GH:omdurman-rules/src/unit.rs:317	is_friendlys	<pre> 315 // "Friendlys" units obey several special rules (§5.21, §5.23, §6.52, 316 // §9.14 victory conditions). 317 pub fn is_friendlys(&self) -> bool { 318 matches!(319 self,</pre>
omdurman-types/src/lib.rs:1015 GH:omdurman-types/src/lib.rs:1015	Friendlys	<pre> 1013 // Native volunteer brigade -- the Shaggyeh (§6.52). Do not receive 1014 // brigade integrity (§5.54 enumerates only British/Egyptian/Sudanese). 1015 Friendlys, 1016 } 1017 }</pre>

Proven by: [omdurman-rules::src::effects::fire::range_table_player_for_routes_friendlys_and_fok_to_the_dervish_table](#)

Covered by tests:

- [omdurman-rules::src::unit_profiles::friendlys_counters_score_by_bank_not_as_leaders](#)
- [omdurman-rules::src::effects::tests::friendlys_validate_and_resolve_on_dervish_table](#)

§6.53 – Royal Engineers demolition

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File	Symbol	Code Snippet
omdurman-rules/src/unit.rs:288 GH:omdurman-rules/src/unit.rs:288	RoyalEngineers	<pre> 286 // The Royal Engineers (§6.53) -- a *specific* 287 // not a class, so we 288 // model it explicitly. 289 RoyalEngineers, 290 }</pre>

omdurman-rules/src/unit.rs:528 GH:omdurman-rules/src/unit.rs:528	demolishing	<pre> 526 /// Set when the Royal Engineers are committed to a 527 /// (§6.53) -- neither offensive fire nor melee 528 pub demolishing: bool, 529 /// Set when a gunboat has lost its engines to a 530 /// river mine (§10.12, roll 530 /// 5-7): it may no longer move under power and instead drifts two hexes per </pre>
omdurman-rules/src/effects/effect.rs:176 GH:omdurman-rules/src/effects/effect.rs:176	Demolition	<pre> 174 175 /// Royal Engineers demolition (rulebook §6.53). 176 Demolition { 177 unit_id: UnitId, 178 target: DemolitionTarget, </pre>
omdurman-rules/src/effects/setup.rs:37 GH:omdurman-rules/src/effects/setup.rs:37	apply_demolition	<pre> 35 /// resolution happens at end of turn via 36 /// ['apply_resolve_demolition'], which 37 /// checks the engineer is still adjacent and 38 /// undisrupted. 39 pub fn apply_demolition(40 state: &mut GameState, 41 unit_id: UnitId, </pre>
omdurman-rules/src/combat.rs:150 GH:omdurman-rules/src/combat.rs:150	DemolitionTarget	<pre> 148 /// disrupted or driven off. 149 #[derive(Serialize, Deserialize, Clone, Copy, 150 PartialEq, Eq, Debug)] 151 pub enum DemolitionTarget { 152 Fort(UnitId), 153 WallHexside(HexsideRef), </pre>
omdurman-rules/src/effects/state/engineering.rs:39 GH:omdurman-rules/src/effects/state/engineering.rs:39	demolition_targets	<pre> 37 /// the rules would accept. Empty when the unit 38 /// doesn't exist or has no 39 /// adjacent target. 40 pub fn demolition_targets(&self, unit_id: UnitId) - 41 > Vec<DemolitionTarget> { 42 let Ok(unit) = self.unit_or_err(unit_id) else { 43 return Vec::new(); </pre>

Covered by tests: [omdurman-rules::src::effects::tests::demolition_requires_engineers_an_adjacent_target_and_a_pending_commit](#)
[omdurman-rules::src::effects::tests::demolition_cancelled_when_engineer_disrupted](#)
[omdurman-rules::src::effects::tests::demolition_cancelled_when_engineer_moved_away](#)
[omdurman-rules::src::effects::tests::demolition_targets_finds_adjacent_fort_and_wall](#)

§6.54 – Forts

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File	Symbol	Code Snippet
omdurman-rules/src/unit.rs:561 GH:omdurman-rules/src/unit.rs:561	ZocReason	<pre> 559 /// Used by the engine when answering "is this hex in 560 /// an enemy ZOC?". 561 #[derive(Serialize, Deserialize, Clone, Copy, 562 PartialEq, Eq, Debug)] 563 pub enum ZocReason { 564 /// Normal ZOC: any non-disrupted unit other than 565 /// an Anglo-Egyptian 566 /// leader (§5.41) projects ZOC into each of its 567 /// six adjacent hexes. </pre>
omdurman-rules/src/unit.rs:568 GH:omdurman-rules/src/unit.rs:568	Fort	<pre> 566 GunboatVsGunboat, 567 /// Forts project ZOC out of, but not into, an 568 /// empty fort (§5.44, §6.54). 569 Fort, 570 /// Walled-city ZOC: extends out through walls and 571 /// gates but not in, 572 /// across a breach in both directions (§5.44). </pre>

omdurman-rules/src/unit.rs:516 GH:omdurman-rules/src/unit.rs:516	UnitState	<pre> 514 /// rather than one big enum. 515 #[derive(Serialize, Deserialize, Clone, Copy, PartialEq, Eq, Hash, Debug, Default)] 516 pub struct UnitState { 517 /// Reference table: "Disrupted units: no ZOC; may not move; may not fire 518 /// offensively or defensively; may not melee; are turned face up at the </pre>
omdurman-rules/src/combat.rs:65 GH:omdurman-rules/src/combat.rs:65	FireAttack	<pre> 63 /// modifiers (rulebook §6). 64 #[derive(Serialize, Deserialize, Clone, Debug, PartialEq, Eq)] 65 pub struct FireAttack { 66 pub firing_player: Player, 67 pub phase: Phase, </pre>
omdurman-rules/src/combat.rs:80 GH:omdurman-rules/src/combat.rs:80	FireAttack::net_modifier	<pre> 78 impl FireAttack { 79 /// Sum of all fire modifiers applied to this attack (rulebook §6.24). 80 pub fn net_modifier(&self) -> i16 { 81 // Saturating fold rather than `sum()`: the modifier list is unbounded 82 // and arrives over the network, so a plain sum can overflow `i16`. </pre>

Covered by tests: `omdurman-rules::src::effects::tests::retreat_before_melee_may_not_land_on_enemy_fort`

§6.61 – Only artillery may fire at gunboats; 3+ to sink

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Only artillery may fire at gunboats. A result of 3 or more on the combat results table is required to sink a gunboat. Any other result is a miss.

File	Symbol	Code Snippet
omdurman-rules/src/unit.rs:167 GH:omdurman-rules/src/unit.rs:167	WeaponClass	<pre> 165 Serialize, Deserialize, Clone, Copy, PartialEq, Eq, PartialOrd, Ord, Hash, Debug, strum::Display, 166)] 167 pub enum WeaponClass { 168 /// Dervish spears and swords -- no ranged fire at all. 169 Melee, </pre>
omdurman-rules/src/effects/state/fire.rs:49 GH:omdurman-rules/src/effects/state/fire.rs:49	GameState::can_fire_at	<pre> 47 /// modifier in the [`FireAttack`] and is responsible for the LOS gate. 48 /// (Howitzer fire ignores LOS entirely -- §6.64.) 49 pub fn can_fire_at(50 &self, 51 firer: UnitId, </pre>

Covered by tests: `omdurman-rules::src::effects::tests::a_gunboat_sunk_by_artillery_scores_victory_points`
`omdurman-rules::src::effects::tests::rifles_may_not_sink_a_gunboat`
`omdurman-rules::src::effects::tests::artillery_sinks_gunboat_only_on_three_plus`

§6.62 – Only artillery may fire at forts; 2+ to destroy

implemented

manual page unknown

Only artillery may fire at forts. A result of 2 or more on the combat results table is required to eliminate a fort. Any other result is a miss. If the fort contains any enemy units at the instant it is destroyed, one unit is eliminated with the fort.

File	Symbol	Code Snippet
omdurman-rules/src/unit.rs:167 GH:omdurman-rules/src/unit.rs:167	WeaponClass	<pre> 165 Serialize, Deserialize, Clone, Copy, PartialEq, Eq, PartialOrd, Ord, Hash, Debug, strum::Display, 166)] 167 pub enum WeaponClass { 168 /// Dervish spears and swords -- no ranged fire at all. 169 Melee,</pre>
omdurman-rules/src/effects/state/fire.rs:49 GH:omdurman-rules/src/effects/state/fire.rs:49	GameState::can_fire_at	<pre> 47 /// modifier in the [`FireAttack`] and is responsible for the LOS gate. 48 /// (Howitzer fire ignores LOS entirely -- §6.64.) 49 pub fn can_fire_at(50 &self, 51 firer: UnitId,</pre>

Covered by tests: [omdurman-rules::src::effects::tests::a_destroyed_fort_takes_one_occupant_with_it_and_scores_it](#)
[omdurman-rules::src::effects::tests::artillery_destroys_fort_on_two_or_better_only](#)
[omdurman-rules::src::effects::tests::destroyed_fort_takes_one_occupant_with_it](#)

§6.63 – Only artillery may breach wall hexsides; 2+ required

implemented

manual page unknown

Only artillery may fire to breach a wall hexside of Khartoum or the walled city of Omdurman. A result of 2 or more on the combat results table is required to breach a wall. Any other result is a miss. The effect of the breach is to negate the wall hexside for line of sight purposes. Place a “BREACH” marker in an adjacent hex so that the arrow points to the breached hexside. If any enemy units are adjacent to the wall hexside at the instant it is breached, one enemy unit is eliminated.

****6.64) Howitzer fire:**** Five units in the game have howitzer fire capability. These are the five named British gunboats. They may fire their artillery factor as direct fire during the Direct Fire Subphase (see 4 and 6.41) and may then fire the same artillery factor as howitzer fire during the Maxim Second Fire and Howitzer Subphase (see 4 and 6.42). Exception: no howitzer fire is allowed during night game turns. To fire howitzer fire, select any target hex between 4 and 10 hexes from the firing gunboat (ignoring the Line of Sight Table) and roll the ten-sided die twice. The first die roll is the Combat Results Table die roll and the second roll is the impact hex die roll. Refer to the Howitzer Fire Subphase on the warbook for the impact hex. The designated target hex is hit on a roll of 7–10. (see manual for full text)

See also: §6.41, §6.42, §6.64

File	Symbol	Code Snippet
omdurman-types/src/lib.rs:198 GH:omdurman-types/src/lib.rs:198	Breach	<pre> 196 /// Breach in a wall (artillery/§6.63 or Royal Engineers/§6.53). ZOC both 197 /// ways; LOS no longer blocked across the hexside. 198 Breach, 199 /// Khor -- gully/wadi. ZOCs do not extend across (\$5.44); advance after 200 /// combat may not cross (§6.82).</pre>
omdurman-rules/src/effects/effect.rs:280 GH:omdurman-rules/src/effects/effect.rs:280	ArtilleryBreachWall	<pre> 278 /// pre-rolled d10 used for the CRT lookup; range/ LOS are re-derived by the 279 /// engine from the firers and `target`. 280 ArtilleryBreachWall { 281 firers: Vec<UnitId>, 282 target: HexsideRef,</pre>
omdurman-rules/src/effects/fire.rs:734 GH:omdurman-rules/src/effects/fire.rs:734	apply_artillery_breach	<pre> 732 /// artillery's CRT roll -- the rulebook specifies the same "2+ required" 733 /// threshold for both trigger styles. 734 pub fn apply_artillery_breach_wall(735 state: &mut GameState, 736 firers: &[UnitId],</pre>

omdurman-rules/src/effects/state/fire.rs:250 GH:omdurman-rules/src/effects/state/fire.rs:250	can_fire_at_wall	<pre> 248 /// range band and resolving the CRT – this method only validates one 249 /// firer at a time. 250 pub fn can_fire_at_wall(251 &self, 252 firer: UnitId,</pre>
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Covered by tests: [omdurman-rules::src::unit_profiles::breach_marker_cell_returns_none](#)
[omdurman-rules::src::effects::tests::artillery_breaches_wall_only_on_crt_two_or_better](#)
[omdurman-rules::src::effects::tests::wall_breach_eliminate_one_adjacent_enemy_unit](#)
[omdurman-rules::src::effects::tests::breach_is_an_opening_for_los_and_effective_kinds](#)

§6.64 – Howitzer fire

implemented

manual page unknown

File	Symbol	Code Snippet
omdurman-rules/src/howitzer_scatter.rs:8 GH:omdurman-rules/src/howitzer_scatter.rs:8	ScatterHexDirection	<pre> 6 /// the diagram away from the firing player. 7 #[derive(serde::Serialize, serde::Deserialize, Clone, Copy, PartialEq, Eq, Hash, Debug)] 8 pub enum ScatterHexDirection { 9 UpperLeft, 10 UpperRight,</pre>
omdurman-rules/src/howitzer_scatter.rs:38 GH:omdurman-rules/src/howitzer_scatter.rs:38	howitzer_scatter	<pre> 36 /// onto a hex-grid offset oriented away from the firer (see 37 /// `GameState::howitzer_impact_hex`). 38 pub fn howitzer_scatter(impact_roll: DieRoll) -> ScatterHexDirection { 39 create::tables_data::SCATTERGRAM[(impact_roll.value() - 1) as usize] 40 }</pre>
omdurman-rules/src/unit.rs:119 GH:omdurman-rules/src/unit.rs:119	GunboatId	<pre> 117 /// fire; "old" gunboats do not (rulebook §2.32). 118 #[derive(Serialize, Deserialize, Clone, Copy, PartialEq, Eq, Hash, Debug, strum::Display)] 119 pub enum GunboatId { 120 /// One of the five new-type named gunboats with howitzer capability. 121 Named(NamedGunboat),</pre>
omdurman-rules/src/effects/effect.rs:89 GH:omdurman-rules/src/effects/effect.rs:89	HowitzerFire	<pre> 87 /// - CRT result applied to units at impact hex (not the original target). 88 /// - Firers marked as fired. 89 HowitzerFire { 90 attack: FireAttack, 91 combat_results_table_roll: DieRoll,</pre>
omdurman-rules/src/effects/fire.rs:18 GH:omdurman-rules/src/effects/fire.rs:18	apply_howitzer_fire	<pre> 16 17 /// Validate and apply a howitzer fire attack (scatter path) (rulebook §6.64). 18 pub fn apply_howitzer_fire(19 state: &mut GameState, 20 attack: &FireAttack,</pre>
omdurman-rules/src/effects/state/fire.rs:49 GH:omdurman-rules/src/effects/state/fire.rs:49	can_fire_at	<pre> 47 /// modifier in the [`FireAttack`] and is responsible for the LOS gate. 48 /// (Howitzer fire ignores LOS entirely -- §6.64.) 49 pub fn can_fire_at(50 &self, 51 firer: UnitId,</pre>

Proven

by: [omdurman-rules::src::effects::opposite_is_an_involution](#) [omdurman-rules::src::effects::step_toward_lands_on_an_adjacent_hex](#)
[omdurman-rules::src::howitzer_scatter::scatter_is_center_exactly_for_rolls_7_to_10](#)

Covered by tests: omdurman-rules::src::howitzer_scatter::howitzer_on_target_7_to_10
omdurman-rules::src::howitzer_scatter::howitzer_scatters_below_7
omdurman-rules::src::howitzer_scatter::howitzer_each_miss_gets_its_printed_hex
omdurman-rules::src::unit_profiles::named_and_old_gunboats_resolve
omdurman-rules::src::effects::tests::named_gunboat_has_howitzer
omdurman-rules::src::effects::tests::a_scattered_howitzer_shell_hits_a_friendly_occupied_hex
omdurman-rules::src::effects::tests::a_rejected_howitzer_shot_leaves_no_impact_record
omdurman-rules::src::effects::tests::named_gunboat_may_fire_howitzer_in_second_subphase
omdurman-rules::src::effects::tests::named_gunboat_direct_fire_uses_artillery_weapon
omdurman-rules::src::effects::tests::named_gunboat_no_howitzer_at_night
omdurman-rules::src::effects::tests::dervish_gunboat_lacks_howitzer

§6.81 – Moving player may fire with all capable units

implicit

manual page unknown

During Offensive Fire phase, the moving player may fire with all of his units capable of firing, up to their maximum range, within the limitations imposed by the rules of combat.

§6.82 – Advance after combat (offensive fire)

implemented

manual page unknown

If an enemy-occupied hex is vacated as a result of offensive fire, friendly units may advance after combat into the vacated hex. To be eligible to advance, the friendly units must have participated in the attack and must have been adjacent to the vacated hex. Note that artillery may not advance, nor may units advance across a wall hexside (except at a gate or breach). Units may never advance after combat across a khor.

File	Symbol	Code Snippet
omdurman-rules/src/effects/effect.rs:163 GH:omdurman-rules/src/effects/effect.rs:163	AdvanceAfterCombat	<pre> 161 /// **Postconditions:** Unit position moved to `to`, `vacated_by_combat` 162 /// entry consumed. 163 AdvanceAfterCombat { unit_id: UnitId, to: HexCoord }, 164 165 // -- Unit state changes ----- </pre>
omdurman-rules/src/effects/dispatch.rs:45 GH:omdurman-rules/src/effects/dispatch.rs:45	apply_advance_after_combat	<pre> 43 } 44 GameEffect::AdvanceAfterCombat { unit_id, to } => { 45 apply_advance_after_combat(state, *unit_id, *to) 46 } 47 GameEffect::RecoverUnit { unit_id } => apply_recover_unit(state, *unit_id), </pre>
omdurman-types/src/lib.rs:241 GH:omdurman-types/src/lib.rs:241	blocks_advance_after_combat	<pre> 239 240 /// Whether advance-after-combat may *not* cross this side (§6.82, §7.6). 241 pub fn blocks_advance_after_combat(self) -> bool { 242 matches!(243 self, </pre>
omdurman-rules/src/effects/state/melee.rs:141 GH:omdurman-rules/src/effects/state/melee.rs:141	can_advance_after_combat	<pre> 139 /// player's unit, not artillery, adjacent to `to`, and `to` now empty. 140 /// Wall/khor hexside restrictions are not enforced (no hexside map data). 141 pub fn can_advance_after_combat(&self, unit_id: UnitId, to: HexCoord) -> Result<(), RuleError> { 142 let unit = self.unit_or_err(unit_id)?; 143 // §6.7: there is no advance after combat as a result of defensive fire. </pre>

Proven by: `omdurman-rules::src::effects::advance_phase_is_atomic` `omdurman-types::src::lib::hexside_blocking_classifiers_are_exact`

Covered by tests: `omdurman-rules::src::effects::tests::can_advance_after_combat_rejects_wall_hexside`

`omdurman-rules::src::effects::tests::can_advance_after_combat_rejects_khor_hexside`

`omdurman-rules::src::effects::tests::advance_requires_combat_vacated_hex`

`omdurman-rules::src::effects::tests::advance_requires_participation`

`omdurman-rules::src::effects::tests::rejected_advance_phase_keeps_vacated_windows`

 7/8 implemented

§7 – Melee Phase

§7 – Melee Phase (chapter)

descriptive

manual page unknown

Melee Phase

§7.1 – Melee strength printed on counter

implemented

manual page unknown

The melee strength of all units is printed on the counter. Note that gunboats have no melee strength. Gunboats may neither melee attack nor be melee attacked.

File	Symbol	Code Snippet
omdurman-rules/src/scalars.rs:42 GH:omdurman-rules/src/scalars.rs:42	MeleeFactor	<pre>40 // Every possible value from the annotated counter set is a named variant. 41 #[derive(Serialize, Deserialize, Clone, Copy, PartialEq, Eq, Hash, Debug, strum::Display)] 42 pub enum MeleeFactor { 43 One = 1, 44 Three = 3,</pre>
omdurman-rules/src/scalars.rs:53 GH:omdurman-rules/src/scalars.rs:53	MeleeFactor::sum	<pre>51 impl MeleeFactor { 52 // Sum multiple melee factors (rulebook §7.1). 53 pub fn sum<'a>(factors: impl IntoIterator<Item = &'a MeleeFactor>) -> u16 { 54 factors.into_iter().map(f f.value()).sum() 55 }</pre>
omdurman-types/src/lib.rs:956 GH:omdurman-types/src/lib.rs:956	may_be_melee_attacked	<pre>954 955 // Gunboats neither attack nor are attacked in melee (§7.1). 956 pub fn may_be_melee_attacked(self) -> bool { 957 !matches!(self, UnitKind::Gunboat { .. }) 958 }</pre>

Proven by: [omdurman-types::src::lib::unit_kind_melee_capability_law_is_exact](#)

Covered by tests: [omdurman-rules::src::effects::tests::melee_rejects_duplicate_attackers_and_partial_defenders](#)
[omdurman-rules::src::effects::tests::single_shot_melee_is_validated_like_a_declaration](#)
[omdurman-rules::src::effects::tests::a_path_may_not_pass_through_an_enemy_hex](#)
[omdurman-rules::src::tests::melee_factor_values_and_sum](#) [omdurman-rules::src::tests::unit_kind_melee_capability](#)

§7.2 – Melee adjacent only, not across wall hexsides

implemented

manual page unknown

Melee simulates the hand-to-hand fighting of the period. Units may melee attack adjacent enemy units only. Units may not melee attack across a wall hexside, but may melee attack through a gate or breach hexside.

File	Symbol	Code Snippet
omdurman-types/src/lib.rs:236	blocks_melee	<pre>234 // Whether melee may *not* be made across this side (§7.2). Gates and 235 // breaches are passable to melee.</pre>

GH:omdurman-types/src/lib.rs:236		<pre> 236 pub fn blocks_melee(self) -> bool { 237 matches!(self, HexsideKind::Wall HexsideKind::ZaribaThornHedge) 238 } </pre>
omdurman-rules/src/effects/state/melee.rs:13 GH:omdurman-rules/src/effects/state/melee.rs:13	can_melee	<pre> 11 /// that may be melee-attacked (gunboats may not -- §7.1), and no wall or 12 /// thorn-hedge hexside blocks the attack (§7.2). 13 pub fn can_melee(&self, attacker: UnitId, defender_hex: HexCoord) -> Result<(), RuleError> { 14 let unit = self.unit_or_err(attacker)?; 15 </pre>

Proven by: [omdurman-types::src::lib::hexside_blocking_classifiers_are_exact](#)

Covered by tests: [omdurman-rules::src::effects::tests::can_melee_gates_phase_adjacency_and_kind](#)
[omdurman-rules::src::effects::tests::can_melee_rejects_wall_hexside](#)
[omdurman-rules::src::effects::tests::can_melee_rejects_thorn_hedge_hexside](#)
[omdurman-rules::src::effects::tests::can_melee_allows_gate_hexside](#)

§7.3 – Simultaneous melee combat

implemented

manual page unknown

Melee combat is considered simultaneous, so that units eliminated by melee attacks still get a melee combat die roll.

File	Symbol	Code Snippet
omdurman-rules/src/effects/effect.rs:112 GH:omdurman-rules/src/effects/effect.rs:112	MeleeCombat	<pre> 110 /// - Winner may advance into vacated hex (§7.6). 111 /// - Victory points awarded for eliminations. 112 MeleeCombat { 113 attack: MeleeAttack, 114 attacker_roll: DieRoll, </pre>
omdurman-rules/src/effects/melee.rs:6 GH:omdurman-rules/src/effects/melee.rs:6	apply_melee_combat	<pre> 4 /// Validated exactly like a declaration ([`validate_melee_attack`]), then 5 /// resolved at once (no §7.5 reaction window). 6 pub fn apply_melee_combat(7 state: &mut GameState, 8 attack: &MeleeAttack, </pre>

Covered by tests: [omdurman-rules::src::effects::tests::melee_resolves_simultaneously](#)

§7.4 – Who may melee attack / defend

implemented

manual page unknown

Only infantry, cavalry, camel units, and Dervish leaders may melee attack. All units (except gunboats — see 7.1) may melee defend.

See also: §7.1

File	Symbol	Code Snippet
omdurman-types/src/lib.rs:945 GH:omdurman-types/src/lib.rs:945	may_melee_attack	<pre> 943 /// Rulebook §7.4 -- only infantry, cavalry, camel and Dervish leaders may 944 /// melee *attack*. All others (except gunboats) may melee *defend* (§7.1). 945 pub fn may_melee_attack(self) -> bool { 946 matches!(947 self, </pre>

omdurman-types/src/lib.rs:882 GH:omdurman-types/src/lib.rs:882	UnitKind	<pre> 880 /// `Some(UnitKind::Marker)` or `None`. 881 #[derive(Serialize, Deserialize, Clone, Copy, PartialEq, Eq, Hash, Debug, strum::Display)] 882 pub enum UnitKind { 883 /// Foot infantry (§2.3): fire / melee / movement. 884 Infantry { </pre>
omdurman-types/src/lib.rs:724 GH:omdurman-types/src/lib.rs:724	DervishTribe	<pre> 722 strum::EnumIter, 723 }) 724 pub enum DervishTribe { 725 Baggara, 726 Jaalin, </pre>
omdurman-rules/src/effects/state/melee.rs:13 GH:omdurman-rules/src/effects/state/melee.rs:13	can_melee	<pre> 11 /// that may be melee-attacked (gunboats may not -- §7.1), and no wall or 12 /// thorn-hedge hexside blocks the attack (§7.2). 13 pub fn can_melee(&self, attacker: UnitId, defender_hex: HexCoord) -> Result<(), RuleError> { 14 let unit = self.unit_or_err(attacker)?; 15 </pre>

Proven by: [omdurman-types::src::lib::unit_kind_melee_capability_law_is_exact](#)

Covered by tests: [omdurman-rules::src::effects::tests::can_melee_gates_phase_adjacency_and_kind](#)

§7.5 – Cavalry/camel retreat before melee

implemented

manual page unknown

Cavalry and camel units may retreat two hexes from an infantry melee attack. Note, however, that only one retreat per unit per turn is permitted. Thus, if their retreat places them adjacent to enemy units whose melee attacks have not yet been resolved, those enemy units may elect to attack the retreating unit(s).

File	Symbol	Code Snippet
omdurman-rules/src/effects/effect.rs:149 GH:omdurman-rules/src/effects/effect.rs:149	RetreatBeforeMelee	<pre> 147 /// 148 /// **Postconditions:** Unit position moved to `to`. 149 RetreatBeforeMelee { unit_id: UnitId, to: HexCoord }, 150 151 /// An attacking unit advances into a hex vacated by combat (rulebook §6.82 </pre>
omdurman-rules/src/effects/dispatch.rs:42 GH:omdurman-rules/src/effects/dispatch.rs:42	apply_retreat_before_melee	<pre> 40 GameEffect::ResolveMelee => apply_resolve_melee(state), 41 GameEffect::RetreatBeforeMelee { unit_id, to } => { 42 apply_retreat_before_melee(state, *unit_id, *to) 43 } 44 GameEffect::AdvanceAfterCombat { unit_id, to } => { </pre>
omdurman-types/src/lib.rs:962 GH:omdurman-types/src/lib.rs:962	may_retreat_before_melee	<pre> 960 /// Cavalry and camel units may retreat two hexes from an infantry melee 961 /// attack (§7.5). 962 pub fn may_retreat_before_melee(self) -> bool { 963 matches!(self, UnitKind::Cavalry { .. } UnitKind::Camel { .. }) 964 } </pre>
omdurman-rules/src/scalars.rs:119 GH:omdurman-	HexDistance	<pre> 117 /// (rulebook §6.22, §7.5). 118 #[derive(Serialize, Deserialize, Clone, Copy, PartialEq, Eq, PartialOrd, Ord, Hash, Debug)] 119 pub struct HexDistance(pub(crate) u16); </pre>

rules/src/ scalars.rs:119		<pre> 120 121 impl HexDistance { </pre>
omdurman- rules/src/effects/ state/melee.rs:72 GH:omdurman- rules/src/effects/ state/melee.rs:72	can_retreat_before_melee	<pre> 70 /// two hexes away and empty. (Does not verify the 71 attacker is infantry -- 72 /// the caller offers the retreat only in response 73 to one.) 74 pub fn can_retreat_before_melee(&self, unit_id: UnitId, to: HexCoord) -> Result<(), RuleError> { 75 let unit = self.unit_or_err(unit_id)?; 76 if !matches!(self.phase, Phase::Melee) { </pre>

Proven by: `omdurman-rules::src::effects::resolve_melee_is_atomic` `omdurman-rules::src::effects::advance_phase_is_atomic`
`omdurman-types::src::lib::unit_kind_melee_capability_law_is_exact`

Covered by tests: `omdurman-rules::src::effects::tests::retreat_before_melee_only_cavalry_two_hexes`
`omdurman-rules::src::effects::tests::retreat_opens_window_only_when_hex_emptyies`
`omdurman-rules::src::effects::tests::rejected_resolve_melee_keeps_the_declaration`

§7.6 – Advance after melee

implemented

manual page unknown

If a melee attack eliminates all of the defenders in an adjacent hex, the Dervish player MUST advance into the vacated hex. To be eligible to advance, the Dervish units must have been adjacent to the vacated hex and participated in the melee attack that eliminated the defenders. All surviving eligible Dervish units MUST advance, up to the stacking limit. The Anglo-Egyptian player may advance if desired. Note that only attacking units may advance.

File	Symbol	Code Snippet
omdurman- rules/src/effects/ effect.rs:163 GH:omdurman- rules/src/effects/ effect.rs:163	AdvanceAfterCombat	<pre> 161 /// **Postconditions:** Unit position moved to 162 `to`; `vacated_by_combat` 163 /// entry consumed. 164 AdvanceAfterCombat { unit_id: UnitId, to: HexCoord }, 165 // -- Unit state changes ----- </pre>
omdurman- rules/src/effects/ dispatch.rs:45 GH:omdurman- rules/src/effects/ dispatch.rs:45	apply_advance_after_combat	<pre> 43 } 44 GameEffect::AdvanceAfterCombat { unit_id, to } => { 45 apply_advance_after_combat(state, *unit_id, *to) 46 } 47 GameEffect::RecoverUnit { unit_id } => apply_recover_unit(state, *unit_id), </pre>
omdurman- rules/src/ef- fects/state/ melee.rs:141 GH:omdurman- rules/src/effects/ state/melee.rs:141	can_advance_after_combat	<pre> 139 /// player's unit, not artillery, adjacent to `to`, 140 and `to` now empty. 141 /// Wall/khor hexside restrictions are not enforced 142 (no hexside map data). 143 pub fn can_advance_after_combat(&self, unit_id: UnitId, to: HexCoord) -> Result<(), RuleError> { 144 let unit = self.unit_or_err(unit_id)?; 145 // §6.7: there is no advance after combat as a 146 result of defensive fire. </pre>
omdurman- rules/src/effects/ melee.rs:6 GH:omdurman- rules/src/effects/ melee.rs:6	apply_melee_combat	<pre> 4 /// Validated exactly like a declaration 5 ([`validate_melee_attack`]), then 6 /// resolved at once (no §7.5 reaction window). 7 pub fn apply_melee_combat(8 state: &mut GameState, 9 attack: &MeleeAttack, </pre>

Proven by: `omdurman-rules::src::effects::advance_phase_is_atomic`

Covered by tests: [omdurman-rules::src::effects::tests::advancing_consumes_the_units_eligibility](#)
[omdurman-rules::src::effects::tests::dervish_advance_after_melee_is_mandatory](#)
[omdurman-rules::src::effects::tests::rejected_advance_phase_keeps_vacated_windows](#)
[omdurman-rules::src::effects::tests::melee_mandatory_advance_leaders_are_free_stacking](#)

§7.7 – Melee modifiers

[implemented](#)

manual page unknown

To resolve melee, both the attacker and the defender roll on the Combat Results Table and apply the applicable melee modifier to their die roll. The Dervish player receives a +2 melee modifier, the Anglo-Egyptian player receives a +1 melee modifier. No terrain modifiers are applied to melee combat (Exception: Zariba hexsides in the historical scenario and the campaign game, if constructed — see 9.23). Melee losses must be taken from meleeing units first!

See also: §9.23

File	Symbol	Code Snippet
omdurman-rules/src/combat.rs:108 GH:omdurman-rules/src/combat.rs:108	MeleeModifier	<pre>106 107 #[derive(Serialize, Deserialize, Clone, Copy, 108 PartialEq, Eq, Debug)] 109 pub enum MeleeModifier { 110 /// +2 to all Dervish melee rolls (§7.7). 111 DervishStandard,</pre>
omdurman-rules/src/effects/melee.rs:402 GH:omdurman-rules/src/effects/melee.rs:402	mandatory_melee_modifier	<pre>400 /// `(attacker_modifiers, defender_modifiers)`; the 401 engine applies exactly 402 these at resolution and rejects a declared attack 403 whose lists differ. 404 pub fn mandatory_melee_modifiers(405 state: &GameState, 406 attack: &MeleeAttack,</pre>
omdurman-rules/src/combat.rs:131 GH:omdurman-rules/src/combat.rs:131	MeleeAttack	<pre>129 /// A melee attack: simultaneous, both sides roll on 130 the Combat Results Table (§7.3, §7.7). 131 #[derive(Serialize, Deserialize, Clone, Debug, 132 PartialEq, Eq)] 133 pub struct MeleeAttack { 134 pub attacker_player: Player, 135 pub attacker_hex: HexCoord,</pre>
omdurman-rules/src/combat.rs:112 GH:omdurman-rules/src/combat.rs:112	MeleeModifier::Anglo-EgyptianStandard	<pre>110 DervishStandard, 111 /// +1 to all Anglo-Egyptian melee rolls (§7.7). 112 AngloEgyptianStandard, 113 /// Inverted to -2 when Dervish units melee-attack 114 across a trench into 115 /// an entrenched defender (§9.232).</pre>
omdurman-rules/src/combat.rs:115 GH:omdurman-rules/src/combat.rs:115	MeleeModifier::DervishVsDefender	<pre>113 /// Inverted to -2 when Dervish units melee-attack 114 across a trench into 115 /// an entrenched defender (§9.232). 116 DervishVsTrenchedDefender, 117 }</pre>
omdurman-rules/src/combat.rs:110 GH:omdurman-rules/src/combat.rs:110	MeleeModifier::DervishStandard	<pre>108 pub enum MeleeModifier { 109 /// +2 to all Dervish melee rolls (§7.7). 110 DervishStandard, 111 /// +1 to all Anglo-Egyptian melee rolls (§7.7). 112 AngloEgyptianStandard,</pre>

Proven by: [omdurman-rules::src::verification::melee_modifier_keeps_roll_legal](#)

Covered by tests: [omdurman-rules::src::effects::tests::melee_resolves_simultaneously](#)
[omdurman-rules::src::effects::tests::melee_modifiers_are_engine_derived_and_mismatches_rejected](#)

2/3 implemented

§8 – Night Game Turns

§8 – Night Game Turns

descriptive

manual page unknown

Night Game Turns

§8.1 – Night effects

implemented

manual page unknown

The effects of night game turns are: a) all Anglo-Egyptian movement allowances are halved (round down), b) there is no Anglo-Egyptian howitzer fire, and c) all fire ranges are halved for both sides (round down, but range 1 stays range 1). Range effects on fire combat are the same as during day game turns. For example, an Anglo-Egyptian infantry unit firing at night will be doubled at range 1, normal at range 2, and may not fire at range 3 or greater.

****8.2) Dervish Desertion Roll:**** Once each campaign game, during the first night turn of the game, the Dervish player rolls one die to see how many of his units desert. The roll is made during the movement phase and the number of deserting Dervish units is equal to 1½ times the roll of one die. The Dervish player may choose which units desert by merely removing them from the mapsheet. The KHALIFA unit, gunboats, artillery units, and forts are the only Dervish units that may not be chosen. No victory points are awarded to the Anglo-Egyptian player for deserting Dervishes.

(see manual for full text)

See also: §8.2

File	Symbol	Code Snippet
omdurman-rules/src/scalars.rs:88 GH:omdurman-rules/src/scalars.rs:88	MovementAllowance::halve	<pre>86 impl MovementAllowance { 87 /// Night movement allowance = halved (round down) (rulebook §8.1, §5.11). 88 pub fn halve(self) -> Self { 89 let v = self.value() / 2; 90 MovementAllowance::try_from(v).expect("halved value always a named variant")</pre>
omdurman-types/src/lib.rs:789 GH:omdurman-types/src/lib.rs:789	DayNight	<pre>787 /// (rulebook §8.1). 788 #[derive(Serialize, Deserialize, Clone, Copy, PartialEq, Eq, Debug)] 789 pub enum DayNight { 790 Day, 791 Night,</pre>
omdurman-rules/src/range_effects.rs:71 GH:omdurman-rules/src/range_effects.rs:71	night_max_range	<pre>69 70 /// The halved maximum range at night (§8.1): round down, minimum 1. 71 pub fn night_max_range(weapon: WeaponClass, ae: bool) - > u8 { 72 let day = max_day_range(weapon, ae); 73 if day <= 1 { 1 } else { day / 2 }</pre>
omdurman-rules/src/scalars.rs:201 GH:omdurman-rules/src/scalars.rs:201	effective_movement_at_night	<pre>199 /// Apply night-turn movement halving for Anglo- Egyptian units (§8.1): all 200 /// Anglo-Egyptian movement allowances are halved (round down). 201 pub fn effective_movement_at_night(202 allowance: MovementAllowance, 203 player: Player,</pre>
omdurman-rules/src/effects/	NoHowitzerAtNight	<pre>110 // §6.64: no howitzer fire at night. 111 if kind == FireKind::Howitzer && self.day_night</pre>

state/fire.rs:112 GH:omdurman-rules/src/effects/state/fire.rs:112		<pre> == DayNight::Night { 112 return Err(RuleError::NoHowitzerAtNight); 113 } 114 </pre>
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Proven by: [omdurman-rules::src::verification::movement_allowance_half_never_panics](#)
[omdurman-rules::src::verification::night_halving_is_ac_only_and_day_neutral](#)
[omdurman-rules::src::range_effects::night_cap_is_halved_day_max_floored_at_one](#)
[omdurman-rules::src::range_effects::night_range_effects_gates_the_day_band_by_the_cap](#)
[omdurman-rules::src::effects::fire::night_capped_distance_is_some_exactly_within_the_night_max](#)

Covered by tests: [omdurman-rules::src::range_effects::night_max_ranges](#) [omdurman-rules::src::range_effects::night_max_ranges_remaining](#)
[omdurman-rules::src::range_effects::ac_rifle_at_night_matches_rulebook_example](#)
[omdurman-rules::src::range_effects::max_day_range_all_combos](#)
[omdurman-rules::src::effects::tests::night_movement_overlay_allowance_halved](#)
[omdurman-rules::src::effects::tests::named_gunboat_no_howitzer_at_night](#)

§8.2 – Dervish Desertion Roll

[implemented](#)

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File	Symbol	Code Snippet
omdurman-rules/src/effects/effect.rs:191 GH:omdurman-rules/src/effects/effect.rs:191	DervishDesertion	<pre> 189 /// the effect. The Khalifa, gunboats, artillery, 190 and forts may not be 191 chosen. 192 DervishDesertion { 193 roll: DieRoll, 194 deserters: Vec<UnitId>, </pre>
omdurman-rules/src/unit.rs:366 GH:omdurman-rules/src/unit.rs:366	is_desertion_exempt	<pre> 364 /// Non-Dervish identities are trivially not 365 eligible to desert and so are 366 reported as exempt too. 367 pub fn is_desertion_exempt(&self) -> bool { 368 match self { 369 UnitIdentity::DervishLeader(DervishLeader::KhalifaAbdullah) </pre>
omdurman-rules/src/turn_track.rs:63 GH:omdurman-rules/src/turn_track.rs:63	DervishDesertion	<pre> 61 None, 62 /// Dervish desertion roll (§8.2) -- occurs on the 63 first night turn. 64 DervishDesertion, 65 /// Dervish reinforcements are available. 66 DervishReinforcements, </pre>
omdurman-rules/src/turn_track.rs:63 GH:omdurman-rules/src/turn_track.rs:63	DervishDesertion	<pre> 61 None, 62 /// Dervish desertion roll (§8.2) -- occurs on the 63 first night turn. 64 DervishDesertion, 65 /// Dervish reinforcements are available. 66 DervishReinforcements, </pre>
omdurman-rules/src/turn_track.rs:60 GH:omdurman-rules/src/turn_track.rs:60	TurnEvent	<pre> 58 /// Special events that occur on specific turns 59 (rulebook §8.2, §9.112, §9.113). 60 #[derive(Serialize, Deserialize, Clone, Copy, 61 PartialEq, Eq, Debug)] 62 pub enum TurnEvent { 63 None, 64 /// Dervish desertion roll (§8.2) -- occurs on the 65 first night turn. </pre>

Proven by: [omdurman-rules::src::effects::desertion_count_is_one_and_a_half_times_the_roll](#)
[omdurman-rules::src::turn_track::desertion_event_is_unique_to_the_first_night_turn](#)

Covered by tests: [omdurman-rules::src::effects::tests::desertion_rejects_a_unit_listed_twice](#)
[omdurman-rules::src::effects::tests::desertion_count_is_floor_one_and_a_half](#)

omdurman-rules::src::turn_track::desertion_on_first_night
omdurman-rules::src::effects::tests::desertion_roll_required_before_first_night_movement_ends
omdurman-rules::src::effects::tests::desertion_removes_chosen_count_and_respects_exemptions

 23/33 implemented

§9 – The Scenarios

§9 – The Scenarios

descriptive

manual page unknown

The Scenarios

§9.1 – The Campaign Game

descriptive

manual page unknown

The Campaign Game

§9.2 – The Historical Scenario

descriptive

manual page unknown

The Historical Scenario

Players should note that the historical scenario is an exercise in futility for the Dervish player. It is, however, an interesting demonstration of the absolute imbecility of the Khalifa's generalship and vividly shows the superiority of entrenched firepower over traditional tribal arms in the colonial period.

§9.3 – Bonus Game: Fall of Khartoum

descriptive

manual page unknown

Bonus Game: FALL OF KHARTOUM

§9.11 – Set Up (Campaign)

descriptive

manual page unknown

Set Up

§9.12 – Scenario Length (Campaign)

implemented

manual page unknown

Scenario Length

6:00 am, Sept. 1 through 8:00 am, Sept. 3, 22 Game Turns.

File	Symbol	Code Snippet
omdurman-rules/src/turn_track.rs:11 GH:omdurman-	GameTime	<pre>9 /// starts at one of these twelve times. 10 #[derive(Serialize, Deserialize, Clone, Copy, PartialEq, Eq, Debug)] 11 pub enum GameTime {</pre>

rules/src/ turn_track.rs:11		12 SixAM, 13 EightAM,
omdurman-rules/src/turn_track.rs:47 GH:omdurman-rules/src/turn_track.rs:47	TurnEntry	45 /// A single entry on the Turn Record Track (rulebook §9.12, §9.22). 46 #[derive(Serialize, Deserialize, Clone, Copy, PartialEq, Eq, Debug)] 47 pub struct TurnEntry { 48 /// 1-based turn number. 49 pub turn: u8,
omdurman-rules/src/turn_track.rs:95 GH:omdurman-rules/src/turn_track.rs:95	CAMPAIGN_TURN_TRACK	93 /// is turn 9, which carries the once-per-game Dervish Desertion Roll (§8.2) -- 94 /// the printed track prints "Dervish Desertion Roll / NIGHT" on that cell. 95 pub const CAMPAIGN_TURN_TRACK: [TurnEntry; 22] = [96 // Row 1, left->right: Sept 1, 6 am -> 8 pm, then the first NIGHT. 97 entry(1, SixAM, Day, TurnEvent::None),
omdurman-rules/src/turn_track.rs:242 GH:omdurman-rules/src/turn_track.rs:242	TurnLabel	240 /// printed cell; `Blank` is for unused positions in the 9×3 grid. 241 #[derive(Serialize, Deserialize, Debug, Clone, Copy, PartialEq, Eq)] 242 pub enum TurnLabel { 243 Blank, 244 /// "SEPT. 1\n6:00 am" -- day header plus time (turn 1).

Proven by: [omdurman-rules::src::turn_track::scenario_turn_indexing_is_exact_for_every_scenario](#)
[omdurman-rules::src::turn_track::turn_labels_agree_with_the_rule_bearing_track](#)

Covered by tests: [omdurman-rules::src::turn_track::campaign_track_22_turns](#) [omdurman-rules::src::turn_track::desertion_on_first_night](#)
[omdurman-rules::src::turn_track::campaign_track_label_and_day_night_agree](#)
[omdurman-rules::src::turn_track::game_time_display_all_variants](#) [omdurman-rules::src::turn_track::turn_label_display](#)
[omdurman-rules::src::turn_track::turn_label_out_of_range_is_none](#)
[omdurman-rules::src::effects::tests::game_over_after_campaign_turns](#)

§9.13 – Special Rules (Campaign)

[descriptive](#)
manual page unknown

- Special Rules
- None.

§9.14 – Victory Conditions (Campaign)

[implemented](#)
manual page unknown

Victory Conditions

The Mahdi’s Tomb in Omdurman was not only the tallest structure in the entire Sudan in 1898, it was also a Dervish holy shrine. Its loss or destruction would be a severe blow to the Mahdist cause. It is accordingly assigned 25 victory points which are awarded to the player who controls it at the conclusion of play. The Dervish player controls it at the start of play. As a tactical note, the Anglo-Egyptian player will find a decisive victory almost impossible unless he takes the Mahdi’s Tomb from the Dervish player. To take the Tomb hex, it must be occupied by one British leader plus any one non-“Friendlies” Anglo-Egyptian combat unit (both undisrupted) at the conclusion of play.

Additional victory points are awarded as follows:

Dervish Player receives:

- 10 pts: each British leader eliminated

(see manual for full text)

See also: §5.21

File	Symbol	Code Snippet
omdurman-rules/src/victory.rs:16 GH:omdurman-rules/src/victory.rs:16	VpSource	<pre> 14 /// the manual and the engine. 15 #[derive(Serialize, Deserialize, Clone, Copy, PartialEq, Eq, Debug)] 16 pub enum VpSource { 17 // ----- Anglo-Egyptian player receives: 18 /// Mahdi's Tomb control at conclusion of play (§9.14). </pre>
omdurman-rules/src/victory.rs:42 GH:omdurman-rules/src/victory.rs:42	VpSource::points	<pre> 40 impl VpSource { 41 /// VP awarded to `who_scores()` (rulebook §9.14). 42 pub fn points(self) -> VictoryPoints { 43 match self { 44 VpSource::MahdisTomb => VictoryPoints::new(25), </pre>
omdurman-rules/src/victory.rs:57 GH:omdurman-rules/src/victory.rs:57	VpSource::who_scores	<pre> 55 56 /// Which player receives these victory points (rulebook §9.14). 57 pub fn who_scores(self) -> Player { 58 match self { 59 VpSource::MahdisTomb </pre>
omdurman-rules/src/victory.rs:92 GH:omdurman-rules/src/victory.rs:92	VictoryLedger	<pre> 90 /// Cumulative victory ledger for one scenario (rulebook §9.14). 91 #[derive(Serialize, Deserialize, Clone, Debug, Default)] 92 pub struct VictoryLedger { 93 pub events: Vec<VpEvent>, 94 } </pre>
omdurman-rules/src/victory.rs:98 GH:omdurman-rules/src/victory.rs:98	VpEvent	<pre> 96 /// A single victory-point scoring event (rulebook §9.14). 97 #[derive(Serialize, Deserialize, Clone, Copy, PartialEq, Eq, Debug)] 98 pub struct VpEvent { 99 pub turn: GameTurnIndex, 100 pub source: VpSource, </pre>
omdurman-rules/src/victory.rs:105 GH:omdurman-rules/src/victory.rs:105	VictoryLedger::total_for	<pre> 103 impl VictoryLedger { 104 /// Total victory points earned by a given player (rulebook §9.14). 105 pub fn total_for(&self, player: Player) -> VictoryPoints { 106 VictoryPoints(107 self.events </pre>
omdurman-rules/src/victory.rs:117 GH:omdurman-rules/src/victory.rs:117	VictoryLedger::superiority	<pre> 115 /// Net superiority: positive = Anglo-Egyptian ahead, negative = Dervish ahead 116 /// (rulebook §9.14). 117 pub fn superiority(&self) -> VictoryPoints { 118 VictoryPoints(119 self.total_for(Player::AngloEgyptian).value() - self.total_for(Player::Dervish).value(), </pre>
omdurman-rules/src/victory.rs:137 GH:omdurman-rules/src/victory.rs:137	CampaignVictoryLevel	<pre> 135 /// Campaign-game victory levels (§9.14). 136 #[derive(Serialize, Deserialize, Clone, Copy, PartialEq, Eq, Debug)] 137 pub enum CampaignVictoryLevel { 138 Draw, 139 Marginal(Player), </pre>
omdurman-rules/src/victory.rs:146 GH:omdurman-rules/src/victory.rs:146	CampaignVictoryLevel::from	<pre> 144 impl CampaignVictoryLevel { 145 /// Assign a level from the net superiority (§9.14). 146 pub fn from_superiority(s: VictoryPoints) -> Self { 147 let net = s.0; 148 // Positive -> Anglo-Egyptian thresholds: 15/30/50 </pre>

omdurman-rules/src/effects/victory.rs:74 GH:omdurman-rules/src/effects/victory.rs:74	score_elimination	<pre>72 /// elimination under `cause`. The owner is derived 73 /// so unlike the historical signature there is no 74 /// caller-supplied player. 75 pub fn score_elimination(state: &mut GameState, 76 unit_id: UnitId, cause: ElimCause) { 77 if let Some(unit) = state.find_unit(unit_id) { 78 let identity = unit.profile.identity;</pre>
omdurman-rules/src/scalars.rs:171 GH:omdurman-rules/src/scalars.rs:171	VictoryPoints	<pre>169 /// (rulebook §9.14). 170 #[derive(Serialize, Deserialize, Clone, Copy, 171 PartialEq, Eq, PartialOrd, Ord, Debug, Default)] 172 pub struct VictoryPoints(pub(crate) i32); 173 impl VictoryPoints {</pre>

Proven by: [omdurman-rules::src::verification::campaign_victory_levels_match_manual_superiority_table](#)
[omdurman-rules::src::verification::vp_source_points_and_scorer_match_the_printed_schedule](#)
[omdurman-rules::src::effects::victory::vp_source_for_routes_every_elimination_to_the_printed_source](#)
[omdurman-rules::src::effects::victory::score_elimination_records_exactly_what_it_scores](#)

Covered by tests: [omdurman-rules::src::effects::tests::a_destroyed_fort_takes_one_occupant_with_it_and_scores_it](#)
[omdurman-rules::src::effects::tests::a_mined_gunboat_scores_and_takes_its_friendlys_down](#)
[omdurman-rules::src::effects::tests::a_gunboat_sunk_by_artillery_scores_victory_points](#)
[omdurman-rules::src::effects::tests::friendlys_bank_scores_by_side](#)
[omdurman-rules::src::effects::tests::mahdis_tomb_not_scored_without_a_leader](#)
[omdurman-rules::src::effects::tests::mahdis_tomb_scores_for_anglo_egyptian_when_held](#)
[omdurman-rules::src::tests::vp_source_attributes](#)

§9.21 – Set Up (Historical)

[descriptive](#)
manual page unknown

Set Up

§9.22 – Scenario Length (Historical)

[implemented](#)
manual page unknown

Scenario Length

6:00 am, September 2 through 12:00 noon, September 2. Four game turns.

File	Symbol	Code Snippet
omdurman-rules/src/turn_track.rs:132 GH:omdurman-rules/src/turn_track.rs:132	HISTORICAL_TURN	<pre>130 131 /// Historical scenario track (§9.22 -- 4 turns, Sept 2 132 6:00 am -> 12:00 pm). 133 pub const HISTORICAL_TURN_TRACK: [TurnEntry; 4] = [134 TurnEntry { 135 turn: 1,</pre>

Proven by: [omdurman-rules::src::turn_track::scenario_turn_indexing_is_exact_for_every_scenario](#)

Covered by tests: [omdurman-rules::src::turn_track::historical_turn_all_four_turns](#)

§9.23 – Special Rule: The Zariba

[descriptive](#)
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Special Rule: “The Zariba”

9.231) Thorn hedge hexsides: -2 to die roll on all Dervish fire attacks; may not melee across in either direction; may not advance after combat across in either direction.

9.232) Trench hexsides: -4 to die roll on all Dervish fire attacks vs. entrenched units only; -2 (instead of +2) melee modifier to Dervish units melee attacking an entrenched unit; entrenched units may be fired “over” in both directions (i.e. they do not block line of sight); units are considered “entrenched” if they are directly adjacent to (and on the Nile River side of) a trench hexside.

(see manual for full text)

See also: §9.231, §9.232

§9.24 – Victory Conditions (Historical)

implemented

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Victory Conditions

Victory Levels are based solely on eliminating enemy units while conserving your own force as much as possible.

| Victory Level | Anglo-Egyptian Player (Dervish units eliminated) | Dervish Player (Anglo-Egyptian units eliminated)
| |—|—|—| | 5 — DECISIVE | 100+ | 30+ | | 4 — STRATEGIC | 60–99 | 15–29 | | 3 — TACTICAL | 45–59 | 10–14 | | 2 — MARGINAL | 30–44 | 5–9 | | 1 — DRAW | 0–29 | 0–4 |

The lower value victory level is then subtracted from the higher level to determine a player’s net victory. For example, if the Anglo-Egyptian player eliminates 104 Dervish units (decisive victory) but loses 18 units doing it (Dervish Strategic), the Anglo-Egyptian player only nets out with a draw (decisive worth 5 minus strategic worth 4 = 1, draw).

(see manual for full text)

File	Symbol	Code Snippet
omdurman-rules/src/victory.rs:176 GH:omdurman-rules/src/victory.rs:176	HistoricalVictoryLevel	<pre>174 /// draw"). 175 #[derive(Serialize, Deserialize, Clone, Copy, PartialEq, Eq, PartialOrd, Ord, Debug)] 176 pub enum HistoricalVictoryLevel { 177 Draw = 1, 178 Marginal = 2,</pre>

Proven by: [omdurman-rules::src::verification::historical_victory_ladders_match_manual_bands](#)

Covered by tests: [omdurman-rules::src::tests::historical_victory_level_for_dervish](#)
[omdurman-rules::src::tests::historical_victory_level_for_anglo_egyptian](#)

§9.31 – Bonus game map

implemented

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Only the small FALL OF KHARTOUM scenario map is used for this game.

File	Symbol	Code Snippet
omdurman-rules/src/board_data.rs:37 GH:omdurman-	fall_of_khartoum_map	<pre>35 36 /// The Fall-of-Khartoum board (§9.3). 37 pub fn fall_of_khartoum_map_data() -> MapData {</pre>

rules/src/ board_data.rs:37		38 FOK.get_or_init(parse(FOK_RON)).clone() 39 }
--------------------------------	--	---

Covered by tests: `omdurman-app::src::tests::scenario_maps_to_board` `omdurman-app::src::tests::start_game_scenario_selects_board`

§9.32 – Set Up (Bonus)

`descriptive`
manual page unknown

Set Up

§9.33 – Scenario Length (Bonus)

`implemented`
manual page unknown

Scenario Length

Variable, see victory conditions (9.35). Rarely lasts five turns.

See also: §9.35

File	Symbol	Code Snippet
omdurman-rules/src/turn_track.rs:172 GH:omdurman-rules/src/turn_track.rs:172	FALL_OF_KHARTOUM	170 /// (the rulebook fixes none); only `day_night` is 171 rule-bearing (night halves 171 /// Anglo-Egyptian movement and ranges and bars 171 howitzer fire, §8.1). 172 pub const FALL_OF_KHARTOUM_TURN_TRACK: [TurnEntry; 8] = 172 [173 TurnEntry { 174 turn: 1,

Proven by: `omdurman-rules::src::turn_track::scenario_turn_indexing_is_exact_for_every_scenario`
`omdurman-rules::src::turn_track::fall_of_khartoum_night_window_is_turns_one_and_two`

Covered by tests: `omdurman-rules::src::turn_track::fall_of_khartoum_turn_one_is_night`
`omdurman-rules::src::turn_track::fall_of_khartoum_turns_3_to_8_are_day`

§9.34 – Special Rules (Bonus)

`descriptive`
manual page unknown

Special Rules

§9.35 – Victory Conditions (Bonus)

`implemented`
manual page unknown

Victory Conditions

Victory is determined by how many turns it takes the Dervish player to eliminate the GORDON leader unit and how many Dervish units are eliminated:

- Dervish decisive: eliminate GORDON turn four or sooner.
- Dervish tactical: eliminate GORDON turn five.

- Dervish marginal: eliminate GORDON turn six.
- British marginal: GORDON survives end of turn six.
- British tactical: GORDON survives end of turn seven.

(see manual for full text)

File	Symbol	Code Snippet
omdurman-rules/src/victory.rs:294 GH:omdurman-rules/src/victory.rs:294	FoKVictoryLevel::resolve	<pre> 292 /// rulebook: GORDON dies turn 5 (tactical) with 24 Dervish losses (-2 293 /// levels) nets a British marginal. 294 pub fn resolve(gordon_died_turn: Option<u8>, scenario_end_turn: u8, dervish_lost: i16) -> Self { 295 let base = Self::base(gordon_died_turn, scenario_end_turn); 296 let base_idx = Self::LADDER </pre>

Proven by: [omdurman-rules::src::verification::fok_victory_ladder_penalties_shift_monotonically](#)

Covered by tests: [omdurman-rules::src::effects::tests::gordon_dying_in_melee_ends_fall_of_khartoum](#)
[omdurman-rules::src::tests::fok_victory_level_worked_example](#) [omdurman-rules::src::tests::fok_victory_level_gordon_died_early](#)
[omdurman-rules::src::tests::fok_victory_level_late_gordon_death](#) [omdurman-rules::src::tests::fok_victory_level_gordon_survived](#)

§9.111 – Dervish set up (Campaign)

implemented

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Dervish player sets up first, moves second.

- Isa Zachneih infantry unit: anywhere on the east bank, in or south of El Debeba.
- KHALIFA ABDULLAH: in the walled city of Omdurman, in either palace hex.
- 3 artillery units, and all Taiasha units: anywhere in the walled city of Omdurman.
- 17 forts: anywhere on the mapsheet south of the Khor Shambat on the west bank, and/or south of all Halfaya hut hexes on the east bank and Nile River islands.
- 2 gunboats: any south edge Nile River hexes.

****9.112) Dervish reinforcements:**** all reinforcements enter on the west edge of the mapsheet, south of the Khor Shambat. Each unit pays the terrain cost of the hex through which it enters, no matter how many units enter through

(see manual for full text)

See also: §9.112

File	Symbol	Code Snippet
omdurman-rules/src/effects/state/setup.rs:105 GH:omdurman-rules/src/effects/state/setup.rs:105	setup_complete	<pre> 103 /// currently shares the same "both sides deployed" gate; when a scenario 104 /// needs a different minimum, branch on `self.scenario` here. 105 pub fn setup_complete(&self) -> Result<(), RuleError> { 106 let has = player { 107 self.units </pre>

Covered by tests: [omdurman-app::src::scenario_setup::campaign_has_no_fixed_placements](#)
[omdurman-rules::src::unit_profiles::hadendowa_first_cell_is_isa_zachneih](#)
[omdurman-rules::src::effects::tests::campaign_deployment_is_boat_land_exclusive](#)
[omdurman-rules::src::effects::tests::campaign_setup_rejects_non_initial_force](#)

§9.112 – Dervish reinforcements (Campaign)

implemented

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File	Symbol	Code Snippet
omdurman-rules/src/effects/effect.rs:183 GH:omdurman-rules/src/effects/effect.rs:183	PlaceReinforcements	<pre> 181 // -- Reinforcement / placement ----- 182 /// Place reinforcements onto the map (rulebook \$9.112, \$9.113). 183 PlaceReinforcements(Vec<UnitPlacement>), 184 185 // -- Scenario-specific ----- </pre>
omdurman-rules/src/effects/dispatch.rs:53 GH:omdurman-rules/src/effects/dispatch.rs:53	apply_place_reinforce	<pre> 51 GameEffect::Demolition { unit_id, target } => apply_demolition(state, *unit_id, *target), 52 GameEffect::PlaceReinforcements(placements) => { 53 apply_place_reinforcements(state, placements) 54 } 55 GameEffect::DervishDesertion { roll, deserters } => { </pre>
omdurman-rules/src/reinforcements.rs:69 GH:omdurman-rules/src/reinforcements.rs:69	dervish_campaign_sched	<pre> 67 /// All reinforcements enter on the west edge, south of the Khor Shambat. 68 /// Each unit pays terrain cost of the hex it enters through. 69 pub fn dervish_campaign_schedule() -> ReinforcementSchedule { 70 ReinforcementSchedule { 71 player: Player::Dervish, </pre>
omdurman-types/src/lib.rs:524 GH:omdurman-types/src/lib.rs:524	Location	<pre> 522 /// Named map landmarks (rulebook mapsheet, \$9.111, \$9.113, \$9.212 scenarios). 523 #[derive(Serialize, Deserialize, Clone, Copy, PartialEq, Eq, Debug, strum::Display)] 524 pub enum Location { 525 FortMakran, 526 NorthFort, </pre>
omdurman-types/src/lib.rs:604 GH:omdurman-types/src/lib.rs:604	SetupLetter	<pre> 602 /// Each letter marks a specific hex where a Dervish leader is placed. 603 #[derive(Serialize, Deserialize, Clone, Copy, PartialEq, Eq, Debug, strum::Display)] 604 pub enum SetupLetter { 605 Y, 606 K, </pre>
omdurman-types/src/lib.rs:747 GH:omdurman-types/src/lib.rs:747	Faction	<pre> 745 /// `Some(BrigadeId::friendlies)`'. 746 #[derive(Serialize, Deserialize, Clone, Copy, PartialEq, Eq, Debug)] 747 pub enum Faction { 748 Dervish { 749 tribe: DervishTribe, </pre>

Proven by: [omdurman-rules::src::reinforcements::wave_for_turn_answers_exactly_on_the_printed_schedule](#)

Covered [omdurman-app::src::tests::place_unit_translates_by_phase](#) by [omdurman-rules::src::reinforcements::dervish_schedule_has_three_waves](#) tests:
[omdurman-rules::src::reinforcements::dervish_wave_one_has_baggaara_and_three_leaders](#)
[omdurman-rules::src::reinforcements::dervish_wave_two_has_hadendowa](#)
[omdurman-rules::src::reinforcements::dervish_wave_three_is_all_remaining](#)
[omdurman-rules::src::reinforcements::wave_for_turn_returns_correct_wave](#)
[omdurman-rules::src::effects::tests::campaign_reinforcements_gate_by_wave](#)
[omdurman-rules::src::effects::tests::campaign_setup_rejects_non_initial_force](#)

\$9.113 – Anglo-Egyptian reinforcements (Campaign)

implemented

manual page unknown

The Anglo-Egyptian player moves first. There are no Anglo-Egyptian units on the mapsheet at start. The GORDON unit is not used in this scenario.

- The leader units KITCHENER, GATACRE, and HUNTER may enter anytime during the first four game turns and do not count against the 12 units per turn limit. All three leaders must be in play by the end of turn four!
- All gunboats enter through any north edge Nile River hex, paying one movement point for the first hex entered. The “Friendlies” brigade enters through the Abu Alim hut hex on the east bank, paying eight movement points per unit. All other Anglo-Egyptian units enter through the west bank “ANGLO-EGYPTIAN ENTRANCE AREA”, each unit paying one movement point to enter the mapsheet.
- Turn 1) Any three gunboats; “Friendlies” brigade; Egyptian Cavalry; Horse Artillery; and two infantry brigades

(see manual for full text)

File	Symbol	Code Snippet
omdurman-rules/src/reinforcements.rs:126 GH:omdurman-rules/src/reinforcements.rs:126	anglo_egyptian_campaign	<pre>124 /// - "Friendlies" enter via Abu Alim hut on the east bank (8 MP per unit). 125 /// - All other AE units enter via the Anglo-Egyptian Entrance Area (1 MP). 126 pub fn anglo_egyptian_campaign_schedule() -> ReinforcementSchedule { 127 let free_leaders = vec![128 CampaignLeader::British(BritishLeader::Kitchener),</pre>

Proven by: [omdurman-rules::src::reinforcements::wave_for_turn_answers_exactly_on_the_printed_schedule](#)

Covered by tests: [omdurman-rules::src::reinforcements::anglo_egyptian_schedule_has_four_waves](#)
[omdurman-rules::src::reinforcements::anglo_egyptian_leaders_available_each_wave](#)
[omdurman-rules::src::reinforcements::anglo_egyptian_turn_four_is_all_remaining](#)
[omdurman-rules::src::effects::tests::campaign_reinforcement_cap_and_double_entry](#)
[omdurman-rules::src::effects::tests::campaign_gunboats_quota_three_per_turn](#)
[omdurman-rules::src::effects::tests::reinforcement_rejected_onto_enemy_occupied_hex](#)
[omdurman-rules::src::effects::tests::campaign_setup_rejects_non_initial_force](#)

§9.211 – Anglo-Egyptian set up first, moves second (Historical)

implemented

manual page unknown

The Anglo-Egyptian player sets up first, and moves second:

- Not in play: GORDON leader unit, “Friendlies” brigade.
- Gunboats start in any Nile River hexes adjacent to the Zariba.
- Camel Corps, Egyptian Cavalry, and Horse Artillery start in the village of Kerreri hut hexes.
- All remaining Anglo-Egyptian units set up in the 13 hexes of the Zariba.

File	Symbol	Code Snippet
omdurman-rules/src/effects/dispatch.rs:454 GH:omdurman-rules/src/effects/dispatch.rs:454	first_player	<pre>452 453 /// The player who moves first in a scenario (\$4, \$9.113, \$9.212, \$9.322). 454 pub fn first_player(scenario: Scenario) -> Player { 455 match scenario { 456 Scenario::Campaign => Player::AngloEgyptian,</pre>

Covered by tests: [omdurman-rules::src::effects::tests::historical_setup_rejects_not_in_play_units](#)

§9.212 – Dervish set up (Historical) – deployment zones and leader-to-hex mapping

implemented

manual page unknown

The Dervish player sets up second, and moves first.

- Not in play: Isa Zachneih, gunboats, and forts.
- All Dervish units must be set up out of the line of sight of all Anglo-Egyptian units.
- Dervish leaders start on the lettered hexes:
 - A: Ali Wad Helu
 - D: Sheik El Din
 - Y: Yakub
 - K: Khalifa Abdullah
 - S: Sherif
 - O: Osman Digna

"A" is the hex of the Dervish player's starting position. "D" is the hex of the Dervish player's starting position. "Y" is the hex of the Dervish player's starting position. "K" is the hex of the Dervish player's starting position. "S" is the hex of the Dervish player's starting position. "O" is the hex of the Dervish player's starting position. (see manual for full text)

File	Symbol	Code Snippet
omdurman-rules/src/effects/state/setup.rs:215 GH:omdurman-rules/src/effects/state/setup.rs:215	in_deployment_zone	<pre>213 /// plan / UI rather than this hex predicate. Documented, not silently 214 /// dropped. 215 pub fn in_deployment_zone(&self, player: Player, hex: HexCoord, is_boat: bool) -> bool { 216 // No board attached -> permissive (unit tests, unbound session). 217 if self.board.terrain.is_empty() {</pre>
omdurman-rules/src/unit.rs:78 GH:omdurman-rules/src/unit.rs:78	DervishLeader::setup_letter	<pre>76 /// (§9.212): A→Ali Wad Helu, D→Sheik El Din, Y→Yakub, K→Khalifa Abdullah, 77 /// S→Sherif, O→Osman Digna. Inverse of [̀dervish_leader_for_setup_letter`]. 78 pub fn setup_letter(self) -> SetupLetter { 79 match self { 80 DervishLeader::AliWadHelu => SetupLetter::A,</pre>
omdurman-rules/src/unit.rs:94 GH:omdurman-rules/src/unit.rs:94	dervish_leader_for_setup	<pre>92 /// inherent impl here, so the mapping is a free function -- the bijective 93 /// inverse of [̀DervishLeader::setup_letter`]. 94 pub fn dervish_leader_for_setup_letter(letter: SetupLetter) -> DervishLeader { 95 match letter { 96 SetupLetter::A => DervishLeader::AliWadHelu,</pre>

Covered by tests: [omdurman-rules::src::effects::tests::deploy_rejected_outside_zone](#)
[omdurman-rules::src::unit_profiles::embedded_leaders_resolve_from_their_host_section](#)
[omdurman-app::src::scenario_setup::historical_places_all_six_leaders_when_anchors_present](#)
[omdurman-app::src::scenario_setup::missing_anchor_is_reported_not_dropped_silently](#)
[omdurman-rules::src::effects::tests::setup_letter_dervish_leader_roundtrip](#)
[omdurman-rules::src::effects::tests::setup_letter_to_dervish_leader_known_values](#)
[omdurman-rules::src::effects::tests::historical_setup_rejects_not_in_play_units](#)

§9.231 – Thorn hedge hexsides

implemented

manual page unknown

File	Symbol	Code Snippet
omdurman-rules/src/combat.rs:28 GH:omdurman-	ZaribaThornHedge	<pre>26 Terrain(i16), 27 /// -2 thorn-hedge defensive modifier (§9.231). 28 ZaribaThornHedge, 29 /// -4 trench defensive modifier (§9.232). Only</pre>

rules/src/ combat.rs:28		<pre> applies vs. "entrenched" 30 /// units (those Nile-side of the trench hexside). </pre>
omdurman-types/src/lib.rs:206 GH:omdurman-types/src/lib.rs:206	ZaribaThornHedge	<pre> 204 Crest, 205 /// Historical-scenario thorn-hedge segment of the Zariba (§9.231). 206 ZaribaThornHedge, 207 /// Historical-scenario trench segment of the Zariba (§9.232). 208 ZaribaTrench, </pre>
omdurman-rules/src/effects/state.rs:311 GH:omdurman-rules/src/effects/state.rs:311	has_zariba_thorn_hedge	<pre> 309 /// *effective* hexsides, so constructed (§5.3) and authored hedges both 310 /// count. 311 pub fn has_zariba_thorn_hedge(&self, hex: HexCoord) -> bool { 312 for n in hex.neighbors() { 313 if self.hexside_effective_is(hex, n, k k == HexsideKind::ZaribaThornHedge) { </pre>

Covered by tests: [omdurman-rules::src::effects::tests::zariba_fire_penalties_apply_to_dervish_fire_only](#)

§9.232 – Trench hexsides

implemented

manual page unknown

File	Symbol	Code Snippet
omdurman-rules/src/combat.rs:31 GH:omdurman-rules/src/combat.rs:31	ZaribaTrenchEntrenched	<pre> 29 /// -4 trench defensive modifier (§9.232). Only applies vs. "entrenched" 30 /// units (those Nile-side of the trench hexside). 31 ZaribaTrenchEntrenched, 32 } 33 </pre>
omdurman-types/src/lib.rs:208 GH:omdurman-types/src/lib.rs:208	ZaribaTrench	<pre> 206 ZaribaThornHedge, 207 /// Historical-scenario trench segment of the Zariba (§9.232). 208 ZaribaTrench, 209 /// One of the two end hexsides of a Zariba trench segment that connect to 210 /// the Nile River (§9.233). Units may only enter/ leave the Zariba via </pre>

Proven by: [omdurman-rules::src::verification::melee_modifier_keeps_roll_legal](#)

Covered by tests: [omdurman-rules::src::effects::tests::trench_entrenched_units_take_trench_modifiers](#)

§9.233 – Zariba entry/exit costs

implemented

manual page unknown

Units may only enter and/or leave the Zariba via the two end hexsides that connect to the Nile River, paying +2 movement points to cross (Exception: advance after combat across an entrenched hexside).

File	Symbol	Code Snippet
omdurman-types/src/lib.rs:258 GH:omdurman-types/src/lib.rs:258	blocks_movement	<pre> 256 /// `omdurman-rules`). The trench *end* variants are therefore intentionally 257 /// not blocking. 258 pub fn blocks_movement(self) -> bool { 259 matches!(260 self, </pre>

omdurman-rules/src/effects/state.rs:324 GH:omdurman-rules/src/effects/state.rs:324	zariba_entry_surcharge	<pre> 322 /// +2 movement points to cross"). Trench ends are authored FoK geography 323 /// (players cannot construct them), so this reads the authored board. 324 pub fn zariba_entry_surcharge(&self, from: HexCoord, to: HexCoord) -> i16 { 325 match self.board.hexside_between(from, to) { 326 Some(k) if k.is_zariba_trench_end() => 2, </pre>
omdurman-rules/src/effects/state/movement.rs:432 GH:omdurman-rules/src/effects/state/movement.rs:432	movement_cost_for	<pre> 430 /// 431 /// \$5.42: entering or leaving an enemy ZOC adds no MP cost. 432 pub fn movement_cost_for(433 &self, 434 unit: &UnitPlacement, </pre>
omdurman-rules/src/effects/movement.rs:33 GH:omdurman-rules/src/effects/movement.rs:33	apply_move_unit	<pre> 31 /// engine from the board (\$5.11/\$5.24): the caller- supplied `cost` is ignored 32 /// (it survives in the effect only for wire compatibility). 33 pub fn apply_move_unit(34 state: &mut GameState, 35 unit_id: UnitId, </pre>

Proven by: [omdurman-types::src::lib::zariba_trench_ends_differ_only_in_the_entry_rule](#)

Covered by tests: [omdurman-rules::src::effects::tests::zariba_end_hexside_costs_extra_mp](#)
[omdurman-rules::src::effects::tests::zariba_thorn_hedge_blocks_movement](#)
[omdurman-app::src::picker::movement::trench_end_crossing_costs_two_extra_mp](#)

§9.321 – British set up (Bonus)

[implemented](#)

manual page unknown

The British player sets up first, moves second:

- General GORDON leader unit in the palace.
- Two old style (unnamed) gunboats in any Nile River hexes.
- Set up in any building or hut hexes of Khartoum, Forts Makran and/or Buri, and/or adjacent to any wall hex:
 - one Egyptian Battalion artillery unit
 - two British infantry units (represents Caucasian troops)
 - three Egyptian infantry units (represents Cairo “Bazouks”)
 - four Sudan infantry units (represents Sudanese blacks)
 - four “Friendlies” units (represents the Shaggyeh)

(see manual for full text)

File	Symbol	Code Snippet
omdurman-rules/src/scenario_setup.rs:42 GH:omdurman-rules/src/scenario_setup.rs:42	FALL_OF_KHARTOUM	<pre> 40 /// North Fort uses a campaign HadendowaForts counter (one of the spare fort 41 /// sprites). 42 pub const FALL_OF_KHARTOUM_SETUP: &[FixedPlacement] = &[43 FixedPlacement { 44 section: SectionName::BritishBoats, </pre>

Covered by tests: [omdurman-rules::src::effects::tests::confirm_ready_rejected_below_scenario_target](#)
[omdurman-rules::src::effects::tests::remove_deployed_unit_happy_path](#)
[omdurman-app::src::scenario_setup::fall_of_khartoum_places_gordon_in_the_palace](#)
[omdurman-app::src::scenario_setup::fall_of_khartoum_reports_missing_palace](#)
[omdurman-app::src::scenario_setup::fall_of_khartoum_fort_landmarks_sit_at_the_correct_hexes](#)

omdurman-rules::src::effects::tests::fok_ae_gunboat_deploys_only_on_nile
omdurman-rules::src::effects::tests::fok_ae_land_unit_rejected_on_nile
omdurman-rules::src::effects::tests::fok_setup_complete_requires_full_oob
omdurman-rules::src::effects::tests::deploy_via_real_sprite_resolution_matches_engine
omdurman-rules::src::effects::tests::british_boats_named_vs_old_gunboat_detection
omdurman-rules::src::effects::tests::fok_order_of_battle_british omdurman-rules::src::effects::tests::fok_order_of_battle_dervish
omdurman-rules::src::effects::tests::fok_caps_bind_across_counter_variants

§9.322 – Dervish enters turn one (Bonus)

implemented

manual page unknown

Dervish player moves first: enters turn one through any hexes on the south or east edge of the map.

- 32 Mulazmin units (represents combined forces of Wad El Nejumi, Abu Girgeh, and Sheik El Obeid)
- 2 Hadendowa; 6 Kehena; 5 Degheim (represents Mahdi's combined west bank forces)
- 3 Dervish artillery units.

File	Symbol	Code Snippet
omdurman-rules/src/unit_profiles.rs:392 GH:omdurman-rules/src/unit_profiles.rs:392	ali_wad_helu	<pre>390 /// §9.2/§9.3) unplaceable -- the FoK order-of-battle table 391 /// (`fok_cap_group`) is keyed by identity. 392 pub fn ali_wad_helu(col: u32, row: u32) -> Option<Classification> { 393 match (col, row) { 394 (0, 0) => dervish_leader(DervishLeader::AliWadHelu),</pre>
omdurman-types/src/lib.rs:846 GH:omdurman-types/src/lib.rs:846	sections_for_picker	<pre>844 /// counter of those two forces and leave the §9.322 entry force -- and 845 /// with it the setup Ready gate (§9.2/§9.3) -- unplaceable. 846 pub fn sections_for_picker(self) -> Option<&'static [SectionName]> { 847 match self { 848 Scenario::Campaign Scenario::Historical => None,</pre>

Covered by tests: omdurman-rules::src::effects::tests::deployment_requires_the_canonical_counter
omdurman-rules::src::effects::tests::fok_reinforcements_need_the_owners_movement_phase_and_unique_ids
omdurman-rules::src::unit_profiles::ali_wad_helu_block_resolves_leader_and_degelim_tribes
omdurman-rules::tests::fok_setup_flow::degheim_counters_resolve_to_the_degheim_tribe
omdurman-rules::src::effects::tests::fok_setup_complete_requires_full_oob
omdurman-rules::src::effects::tests::fok_dervish_land_unit_rejected_on_nile
omdurman-types::src::lib::fok_picker_allowlist_has_dervish_entry_force_blocks
omdurman-rules::src::effects::tests::fok_order_of_battle_dervish
omdurman-rules::src::effects::tests::fok_dervish_east_edge_on_diamond_board

§9.341 – Turn 1 is always a night turn (Bonus)

implemented

manual page unknown

Turn 1 is always a night turn (see 8.1).

See also: §8.1

File	Symbol	Code Snippet
omdurman-rules/src/turn_track.rs:172	FALL_OF_KHARTOUM	<pre>170 /// (the rulebook fixes none); only `day_night` is rule-bearing (night halves</pre>

GH:omdurman-rules/src/turn_track.rs:172		<pre>171 /// Anglo-Egyptian movement and ranges and bars howitzer fire, §8.1). 172 pub const FALL_OF_KHARTOUM_TURN_TRACK: [TurnEntry; 8] = [173 TurnEntry { 174 turn: 1,</pre>
---	--	---

Proven by: `omdurman-rules::src::turn_track::scenario_turn_indexing_is_exact_for_every_scenario`
`omdurman-rules::src::turn_track::fall_of_khartoum_night_window_is_turns_one_and_two`

Covered by tests: `omdurman-rules::src::turn_track::fall_of_khartoum_turn_one_is_night`

§9.342 – All hexes are playable, including half hexes (Bonus)

implemented

manual page unknown

All hexes are playable, including hexes showing up half or less.

File	Symbol	Code Snippet
<code>omdurman-rules/src/board_data.rs:37</code> GH:omdurman-rules/src/board_data.rs:37	<code>fall_of_khartoum_map</code>	<pre>35 36 /// The Fall-of-Khartoum board (§9.3). 37 pub fn fall_of_khartoum_map_data() -> MapData { 38 FOK.get_or_init(parse(FOK_RON)).clone() 39 }</pre>

Covered by tests: `omdurman-rules::src::effects::tests::fall_of_khartoum_board_excludes_no_hexes`

§9.343 – Both players use the Dervish Range Effects Table (Bonus)

implemented

manual page unknown

Both players must use the Dervish Range Effects Table.

File	Symbol	Code Snippet
<code>omdurman-rules/src/effects/fire.rs:78</code> GH:omdurman-rules/src/effects/fire.rs:78	<code>range_band_for</code>	<pre>76 /// in FALL OF KHARTOUM *both* players use the Dervish Range Effects Table 77 /// (§9.343). 78 pub fn range_band_for(79 scenario: Scenario, 80 player: Player,</pre>

Proven by: `omdurman-rules::src::effects::fire::range_band_for_routes_to_the_right_faction_table`
`omdurman-rules::src::effects::fire::range_table_player_for_routes_friendlies_and_fok_to_the_dervish_table`

Covered by tests: `omdurman-rules::src::effects::tests::fok_both_players_use_dervish_range_table`

§9.344 – Dervish controls the North Fort (Bonus)

implemented

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The Dervish player controls the “North Fort” and may fire its guns.

File	Symbol	Code Snippet
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omdurman-rules/src/scenario_setup.rs:42 GH:omdurman-rules/src/scenario_setup.rs:42	FALL_OF_KHARTOUM	<pre> 40 /// North Fort uses a campaign HadendowaForts counter (one of the spare fort 41 /// sprites). 42 pub const FALL_OF_KHARTOUM_SETUP: &[FixedPlacement] = &[43 FixedPlacement { 44 section: SectionName::BritishBoats,</pre>
omdurman-rules/src/effects/state.rs:413 GH:omdurman-rules/src/effects/state.rs:413	hex_has_enemy_fort	<pre> 411 /// may neither occupy an enemy fort nor advance after combat into one 412 /// (forts are never captured -- only destroyed, \$6.62/\$6.53/\$7.6). 413 pub fn hex_has_enemy_fort(&self, hex: HexCoord, mover: Player) -> bool { 414 self.units.iter().any(u { 415 u.position == hex</pre>

Covered by tests: [omdurman-app::src::scenario_setup::fall_of_khartoum_places_gordon_in_the_palace](#)
[omdurman-app::src::scenario_setup::fall_of_khartoum_fort_landmarks_sit_at_the_correct_hexes](#)
[omdurman-app::src::scenario_setup::placement_done_gate_matches_by_identity_not_allocated_id](#)
[omdurman-rules::src::effects::tests::fok_order_of_battle_dervish](#)
[omdurman-types::src::lib::fok_picker_allowlist_has_dervish_entry_force_blocks](#)

\$9.345 – Gunboat White Nile <-> Blue Nile crossing (Bonus)

implemented

manual page unknown

The British gunboats may move from the White Nile to the Blue Nile and vice-versa at an off-board movement cost of six “upstream” movement points.

File	Symbol	Code Snippet
omdurman-rules/src/effects/state.rs:397 GH:omdurman-rules/src/effects/state.rs:397	is_nile_mouth_crossing	<pre> 395 /// must be named on the board, else this is `false` and the move falls 396 /// through to the ordinary contiguous-Nile rules. 397 pub fn is_nile_mouth_crossing(&self, from: HexCoord, to: HexCoord) -> bool { 398 let white = self 399 .board</pre>
omdurman-types/src/lib.rs:540 GH:omdurman-types/src/lib.rs:540	Location::WhiteNileMouth	<pre> 538 /// The off-board mouth of the White Nile branch (FALL OF KHARTOUM \$9.345) -- 539 /// a British gunboat may cross to the Blue Nile mouth for 6 upstream MP. 540 WhiteNileMouth, 541 /// The off-board mouth of the Blue Nile branch (FALL OF KHARTOUM \$9.345). 542 BlueNileMouth,</pre>

Covered by tests: [omdurman-rules::src::effects::tests::fok_gunboat_crosses_between_nile_mouths](#)

\$9.346 – GORDON immobile, eliminated only at the Palace (Bonus)

implemented

manual page unknown

The GORDON leader unit starts in the palace and may not move during the scenario. He may only be eliminated by a Dervish unit passing through or occupying the palace hex (as normal movement or as advance after combat).

File	Symbol	Code Snippet
omdurman-rules/src/effects/	check_gordon_palace	<pre> 590 /// (overrun in passing, \$6.51). A no-op outside FoK or while he lives</pre>

dispatch.rs:592 GH:omdurman-rules/src/effects/dispatch.rs:592		<pre>591 /// unthreatened. 592 pub fn check_gordon_palace(state: &mut GameState) { 593 if state.scenario != Scenario::FallOfKhartoum { 594 return;</pre>
omdurman-rules/src/unit.rs:332 GH:omdurman-rules/src/unit.rs:332	UnitIdentity::is_gordon	<pre>330 /// Whether this is the GORDON leader unit (\$9.32, 331 /// palace defender whose elimination ends FALL OF 332 /// KHARTOUM (\$9.35). 333 pub fn is_gordon(&self) -> bool { 334 matches!(self,</pre>

Covered by tests: [omdurman-rules::src::unit_profiles::gordon_is_an_immobile_british_leader](#)
[omdurman-rules::src::effects::tests::gordon_survives_means_no_elimination](#)
[omdurman-rules::src::effects::tests::gordon_dying_in_melee_ends_fall_of_khartoum](#)
[omdurman-app::src::scenario_setup::fall_of_khartoum_places_gordon_in_the_palace](#)

  7/10 implemented

§10 – Optional Rules

§10 – Optional Rules

descriptive

manual page unknown

Optional Rules (Campaign game only)

It is suggested that the most intriguing employment of the following two options is to permit the Dervish player to have either one or the other, but the Anglo-Egyptian player doesn't know which one until he stumbles onto it. Players are advised that the employment of both optionals in the same game is not recommended.

§10.1 – River Mines

descriptive

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River Mines

The Khalifa twice tried (unsuccessfully) to submerge a powerful mine in the Nile to sink or damage British gunboats. This option assumes that both attempts were successful.

§10.2 – River Chain

descriptive

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River Chain

The Khalifa also tried (also unsuccessfully) to string a heavy chain across the Nile to stop or slow down the British gunboats. This option assumes the chain was emplaced.

§10.11 – Secretly record mine hexes

implemented

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Prior to the commencement of play the Dervish player secretly records two Nile River hexes to be mined (the mines may not both be placed in the same hex). These hexes must be south of the E–W hexrow in which the Khor Shambat empties into the Nile.

File	Symbol	Code Snippet
omdurman-rules/src/effects/setup.rs:354 GH:omdurman-rules/src/effects/setup.rs:354	apply_place_mine	<pre>352 /// Lay a river mine during setup (§10.11). Validated by 353 /// [<code>GameState::can_place_mine`</code>]. 354 pub fn apply_place_mine(state: &mut GameState, hex: HexCoord) -> Result<(), RuleError> { 355 state.can_place_mine(hex)?; 356 state.mines.push(MinePlacement {</pre>
omdurman-rules/src/effects/state.rs:205 GH:omdurman-rules/src/effects/state.rs:205	GameState::mines	<pre>203 204 friendlies_transport: None, optional_rules: Vec::new(), 205 mines: Vec::new(), chain: None, 206 board: Arc::new(BoardInfo::default()),</pre>
omdurman-rules/src/	OptionalRule	<pre>56 /// two should be in play (rulebook §10). 57 #[derive(Serialize, Deserialize, Clone, Copy,</pre>

turn.rs:58 GH:omdurman-rules/src/turn.rs:58		PartialEq, Eq, Debug)) 58 pub enum OptionalRule { 59 RiverMines, 60 RiverChain,
--	--	--

Covered by tests: [omdurman-rules::src::effects::tests::mine_and_chain_limits_enforced_in_setup](#)
[omdurman-rules::src::effects::tests::mines_and_chain_require_their_optional_rule](#)

§10.12 – Mine resolution

implemented

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When a British gunboat enters a mined hex, the Dervish player must order it to stop as it has struck a mine. The Dervish player then resolves the effect of the mine's blast by rolling the ten-sided die:

- 1–4: No effect
- 5–7: Gunboat damaged, lost use of its engines and must drift two hexes per turn (with the current) for the rest of the game. No effect on guns or Maxims unless they drift out of range.
- 8–10: Gunboat sunk!

File	Symbol	Code Snippet
omdurman-rules/src/effects/effect.rs:201 GH:omdurman-rules/src/effects/effect.rs:201	RiverMine	199 // -- Optional rules ----- 200 /// River mine resolution (rulebook §10.12). 201 RiverMine { 202 gunboat_id: UnitId, 203 hex: HexCoord,
omdurman-rules/src/effects/river.rs:128 GH:omdurman-rules/src/effects/river.rs:128	apply_river_mine	126 127 /// Apply a river-mine resolution (rulebook §10.12). 128 pub fn apply_river_mine(129 state: &mut GameState, 130 gunboat_id: UnitId,
omdurman-rules/src/transport.rs:63 GH:omdurman-rules/src/transport.rs:63	MineResult	61 /// British gunboat enters a mined hex. 62 #[derive(Serialize, Deserialize, Clone, Copy, PartialEq, Eq, Debug)] 63 pub enum MineResult { 64 /// Roll 1-4: no effect. 65 NoEffect,

Proven by: [omdurman-rules::src::verification::mine_result_bands_match_the_printed_rule](#)
[omdurman-rules::src::effects::river::river_mine_triggers_even_on_a_harmless_roll](#)
[omdurman-rules::src::effects::river::river_mine_engine_loss_armes_the_drift_flag](#)
[omdurman-rules::src::effects::river::river_mine_sinking_removes_the_gunboat](#)

Covered by tests: [omdurman-rules::src::effects::tests::a_mined_gunboat_scores_and_takes_its_friendlies_down](#)
[omdurman-rules::src::effects::tests::mine_fires_once_and_sparees_dervish](#)

§10.13 – Mines consumed after both rolled for

implemented

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After both mines have been rolled for, no more are available.

File	Symbol	Code Snippet
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omdurman-rules/src/effects/river.rs:128 GH:omdurman-rules/src/effects/river.rs:128	apply_river_mine	<pre> 126 127 /// Apply a river-mine resolution (rulebook §10.12). 128 pub fn apply_river_mine(129 state: &mut GameState, 130 gunboat_id: UnitId, </pre>
---	-------------------------	---

Proven by: [omdurman-rules::src::effects::river::river_mine_triggers_even_on_a_harmless_roll](#)
[omdurman-rules::src::effects::river::a_triggered_mine_never_fires_again](#)

Covered by tests: [omdurman-rules::src::effects::tests::mine_fires_once_and_spares_dervish](#)

§10.14 – Dervish gunboats pass safely

implemented

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The Dervish player’s gunboats may pass through the mined hexes with no ill effect (he knows where the mines are).

File	Symbol	Code Snippet
omdurman-rules/src/effects/river.rs:128 GH:omdurman-rules/src/effects/river.rs:128	apply_river_mine	<pre> 126 127 /// Apply a river-mine resolution (rulebook §10.12). 128 pub fn apply_river_mine(129 state: &mut GameState, 130 gunboat_id: UnitId, </pre>

Proven by: [omdurman-rules::src::effects::river::dervish_gunboats_pass_mined_hexes_unharmed](#)

Covered by tests: [omdurman-rules::src::effects::tests::mine_fires_once_and_spares_dervish](#)

§10.21 – Secretly record chain hexes

implemented

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Prior to the commencement of play the Dervish player secretly records a line of river hexes (not exceeding four hexes long) across which the chain is strung. The hexes must be south of the E–W hexrow in which the Khor Shambat empties into the Nile.

File	Symbol	Code Snippet
omdurman-rules/src/effects/state/setup.rs:481 GH:omdurman-rules/src/effects/state/setup.rs:481	can_place_chain	<pre> 479 /// Read-only check of a river-chain placement in 480 setup (§10.21): Setup phase 481 /// and at most [<code>MAX_CHAIN_HEXES`</code>] hexes. 482 pub fn can_place_chain(&self, hexes: &[HexCoord]) - 483 > Result<(), RuleError> { 484 self.require_setup_phase()?; 485 // Optional-rule gate: the chain exists only 486 when the River Chain option </pre>
omdurman-rules/src/effects/state/setup.rs:365 GH:omdurman-rules/src/effects/state/setup.rs:365	apply_place_chain	<pre> 363 /// Lay (or replace) the river chain during setup 364 (§10.21). Validated by 365 /// [<code>GameState::can_place_chain`</code>]. 366 pub fn apply_place_chain(state: &mut GameState, hexes: 367 &[HexCoord]) -> Result<(), RuleError> { 368 state.can_place_chain(hexes)?; 369 state.chain = Some(ChainPlacement { </pre>
omdurman-rules/src/effects/state/setup.rs:91 GH:omdurman-	MAX_CHAIN_HEXES	<pre> 89 90 /// Maximum contiguous Nile hexes the river chain may 91 span (§10.21). 92 pub const MAX_CHAIN_HEXES: usize = 4; </pre>

rules/src/effects/ state/setup.rs:91		92 93 <code>impl GameState {</code>
---	--	--

Covered by tests: `omdurman-rules::src::effects::tests::mine_and_chain_limits_enforced_in_setup`
`omdurman-rules::src::effects::tests::mines_and_chain_require_their_optional_rule`

§10.22 – Gunboat stops on chained hex

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When a British gunboat enters a “chained” river hex it must stop and may move no further that turn.

File	Symbol	Code Snippet
omdurman-rules/src/effects/state/movement.rs:463 GH:omdurman-rules/src/effects/state/movement.rs:463	<code>can_move_gunboat</code>	461 <code>/// The engine computes the cost (one MP per hex, a flat 6 for the \$9.345</code> 462 <code>/// mouth crossing); a larger caller `cost` must also fit.</code> 463 <code>pub fn can_move_gunboat(</code> 464 <code>&self,</code> 465 <code>unit_id: UnitId,</code>

Covered by tests: `omdurman-rules::src::effects::tests::chain_stops_gunboat_until_sunk`

§10.23 – Sinking the chain

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No gunboats (British or Dervish) may cross the chain until it has been sunk by the British player. He may sink the chain by a) having an infantry or cavalry unit spend one complete turn on either riverbank adjacent to a “chained” river hex, or b) firing at the chain with artillery and achieving a 3 or more on the Combat Results Table.

File	Symbol	Code Snippet
omdurman-rules/src/effects/river.rs:173 GH:omdurman-rules/src/effects/river.rs:173	<code>apply_sink_chain</code>	171 <code>/// Sink the river chain (rulebook §10.23). Marks the placed chain cleared so it</code> 172 <code>/// no longer stops gunboats (§10.22).</code> 173 <code>pub fn apply_sink_chain(state: &mut GameState) -></code> 174 <code>Result<(), RuleError> {</code> 175 <code>match state.chain.as_mut() {</code> <code>Some(chain) if !chain.sunk => {</code>

Covered by tests: `omdurman-rules::src::effects::tests::chain_stops_gunboat_until_sunk`

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§11 – Historical Notes

§11 – Historical Notes

descriptive

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Historical Notes

In 1881 Mohammed Ahmed Ibn Al-Sayid Abdullah, the son of an obscure carpenter in the hinterlands of the Sudan, proclaimed himself the “Mahdi” — the Messiah of the Islamic faith. His timing was propitious indeed. Since the early 1820’s a corrupt Egypt, with the Sultan of Turkey’s blessing, had incessantly raped the Sudan, taking ivory and flooding the slave markets with some half million captured Sudanese blacks. By 1880, nearly 40,000 Egyptian troops occupied outposts scattered throughout the Sudan, enforcing Egypt’s hold on this lucrative ivory and slave trade and squeezing the native population dry through vicious and corrupt tax officials. All was controlled from Khartoum via the office of Governor General of the Sudan. The title had been held by a succession of individuals, including General Charles Gordon, whose appointment was an attempt to reinstate some rudimentary justice in the Sudan after France and Britain assumed joint political control of a bankrupt Egypt.

By 1881, however, Gordon’s term had expired and a new Governor General, again corrupt and incompetent,
(see manual for full text)

See also: §9.1, §9.2, §9.3

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Credits

§Credits – Credits

[descriptive](#)

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Combat Results Table (shared reference)

§CRT – Combat Results Table (shared by §6.22 fire and §7.7 melee)

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File	Symbol	Code Snippet
omdurman-rules/src/combat_results_table.rs:10 GH:omdurman-rules/src/combat_results_table.rs:10	FireFactorRow	<pre>8 serde::Serialize, serde::Deserialize, Clone, Copy, 9 PartialEq, Eq, PartialOrd, Ord, Hash, Debug, 10 } 11 pub enum FireFactorRow { 12 /// 1-5 factors 13 Row01to05,</pre>
omdurman-rules/src/combat_results_table.rs:46 GH:omdurman-rules/src/combat_results_table.rs:46	from_total	<pre>44 /// Determine which row a given total fire factor 45 falls into (rulebook §6.22). 46 pub fn from_total(total: u16) -> Self { 47 match total { 48 0..=5 => FireFactorRow::Row01to05,</pre>
omdurman-rules/src/combat_results_table.rs:88 GH:omdurman-rules/src/combat_results_table.rs:88	combat_results_table	<pre>86 /// D = `Disrupt` (1/2 of target units, round up) 87 /// 1...5 = `Eliminate(n)` (that many units removed) 88 pub fn combat_results_table(row: FireFactorRow, roll: 89 DieRoll) -> CombatResult { 90 crate::tables_data::CRT[row.index()][roll.value() 91 - 1] as usize]</pre>
omdurman-rules/src/combat.rs:97 GH:omdurman-rules/src/combat.rs:97	CombatResult	<pre>95 /// * `--` -- no effect 96 #[derive(Serialize, Deserialize, Clone, Copy, 97 PartialEq, Eq, Debug)] 98 pub enum CombatResult { 99 NoEffect, 100 Disrupt,</pre>

Proven by: [omdurman-rules::src::verification::disrupt_half_is_rounded_up](#)
[omdurman-rules::src::combat_results_table::crt_eliminate_count_stays_within_printed_bounds](#)
[omdurman-rules::src::combat_results_table::crt_is_monotone_in_the_die_roll_for_every_row](#)

Covered by tests: [omdurman-rules::src::combat_results_table::ae_combat_results_table_lowest_is_no_effect](#)
[omdurman-rules::src::combat_results_table::ae_combat_results_table_highest_is_eliminate_5](#)
[omdurman-rules::src::combat_results_table::ae_combat_results_table_progresses_with_roll](#)
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[omdurman-rules::src::combat_results_table::fire_factor_row_index_sequential](#)
[omdurman-rules::src::combat_results_table::crt_all_rows_monotone_non_decreasing](#)
[omdurman-rules::src::combat_results_table::crt_cross_row_monotone_for_each_roll](#)
[omdurman-rules::src::combat_results_table::crt_lowest_row_is_worst_highest_row_is_best](#)
[omdurman-rules::src::combat_results_table::crt_eliminate_never_exceeds_5](#)
[omdurman-rules::src::combat_results_table::crt_every_cell_matches_the_table](#)

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| Reference – Charts and Tables

\$Reference – Charts and Tables

out-of-scope

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